

Federal Resource Allocation Efficiency Audit

A Systems Analysis of Recoverable Misallocation in U.S. Federal Spending

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Abstract

This report applies systems engineering methodology to quantify allocative inefficiency in U.S. federal spending. We define government efficiency as the rate of converting fiscal inputs into two terminal outcomes: after-tax real median income growth and median healthy life years. Analysis reveals the U.S. system operates at 52-67% efficiency versus 75-85% for OECD benchmark nations. Using Monte Carlo simulation across five subsystems (defense, healthcare, justice, regulatory/tax, and subsidies), we estimate an Aggregate Efficiency Gap of \$1.85-3.47 trillion annually (P5-P95 range, mean \$2.27T). Closing this gap to OECD median levels would recover approximately \$1.5 trillion per year. The economic equivalent of current inefficiency equals 22.7 million QALYs or 165,700 VSL-equivalents annually.

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1 Executive Summary

This audit applies engineering systems analysis to federal resource allocation. The core finding:

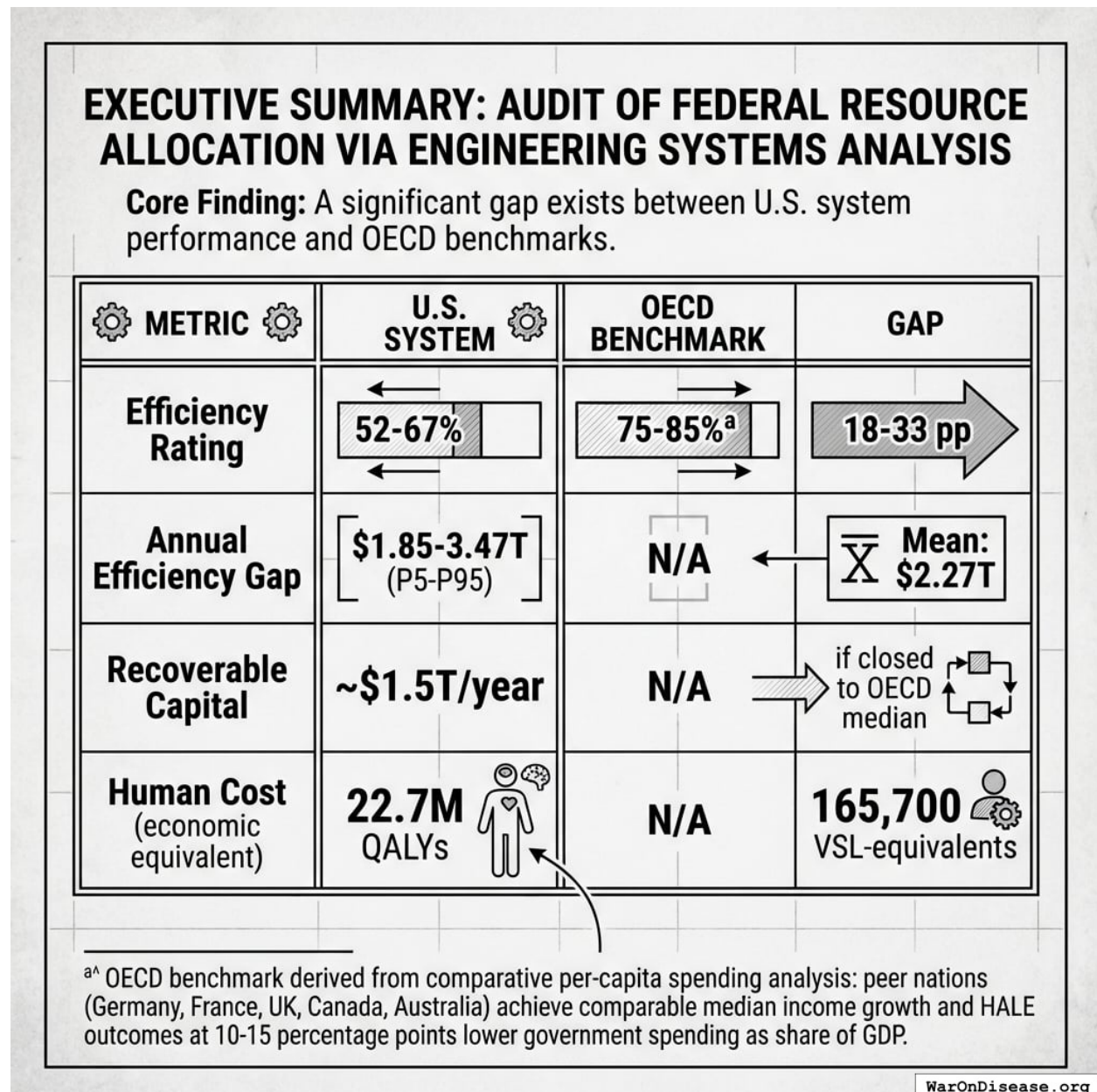


Figure 1: A comparative bar chart showing the efficiency rating gap between the U.S. and OECD benchmarks, highlighting the \$2.27 trillion annual efficiency gap and the equivalent human cost in quality-adjusted life years.

Metric	U.S. System	OECD Benchmark	Gap
Efficiency Rating	52-67%	75-85% ^a	18-33 pp

Metric	U.S. System	OECD Benchmark	Gap
Annual Efficiency Gap	\$1.85-3.47T (P5-P95)	N/A	Mean: \$2.27T
Recoverable Capital	~\$1.5T/year	N/A	if closed to OECD median
Human Cost (economic equivalent)	22.7M QALYs	N/A	165,700 VSL-equivalents

^a *OECD benchmark derived from comparative per-capita spending analysis: peer nations (Germany, France, UK, Canada, Australia) achieve comparable median income growth and HALE outcomes at 10-15 percentage points lower government spending as share of GDP.*

Interpretation Note

The “human cost” figures are **economic equivalents**, not epidemiological mortality counts. Dividing the efficiency gap by VSL (\$13.7M) or QALY threshold (\$100K) yields a measure of foregone welfare, not literal deaths prevented.

The efficiency gap represents capital that could fund the 1% Treaty (\$27.2B/year) 55 times over, or 10x current global disease R&D spending.

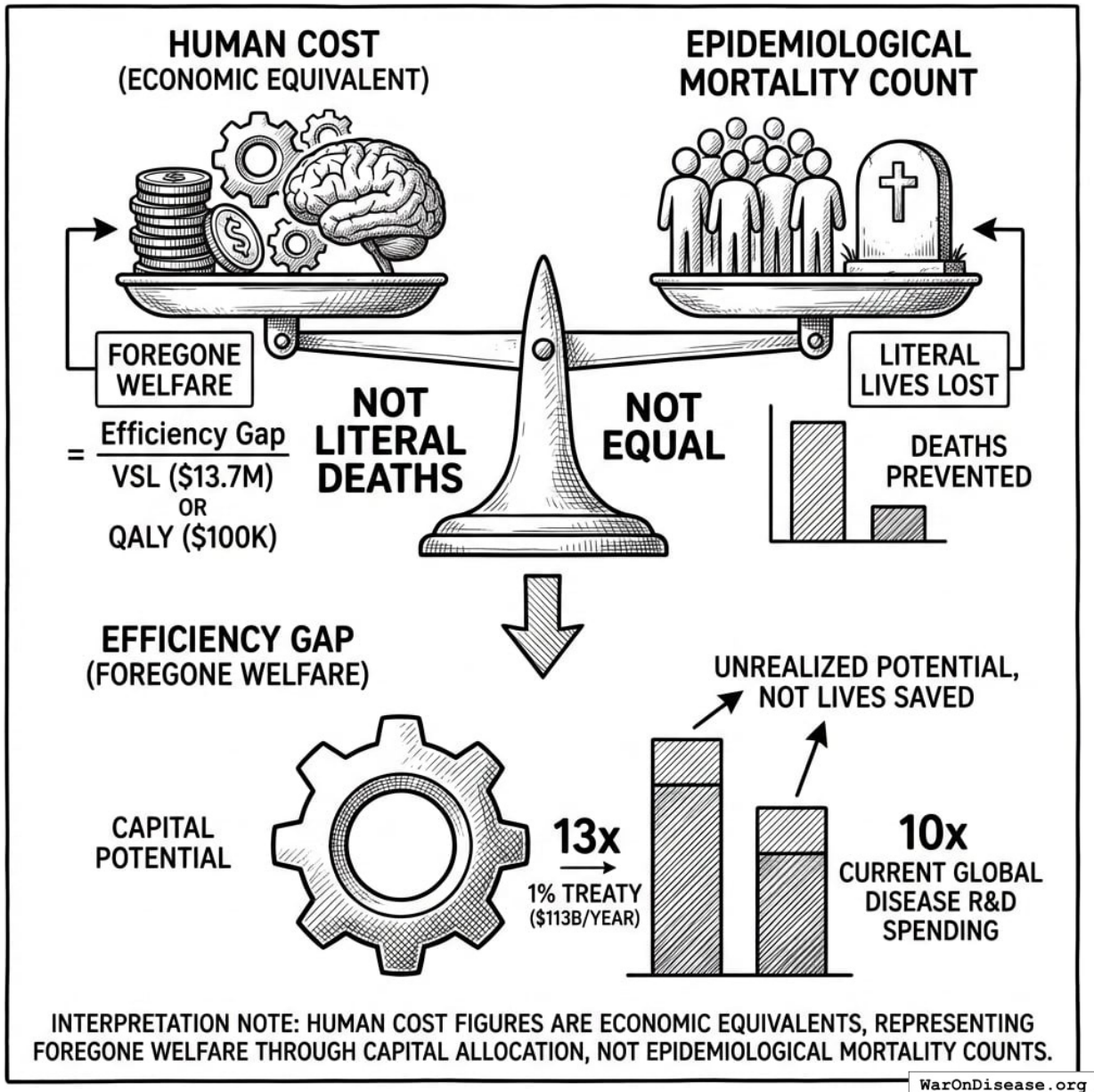


Figure 2: A proportional comparison chart illustrating the magnitude of the efficiency gap relative to the \$27B 1% Treaty annual funding and current global disease R&D spending.

2 System Specifications

2.1 Designed Function

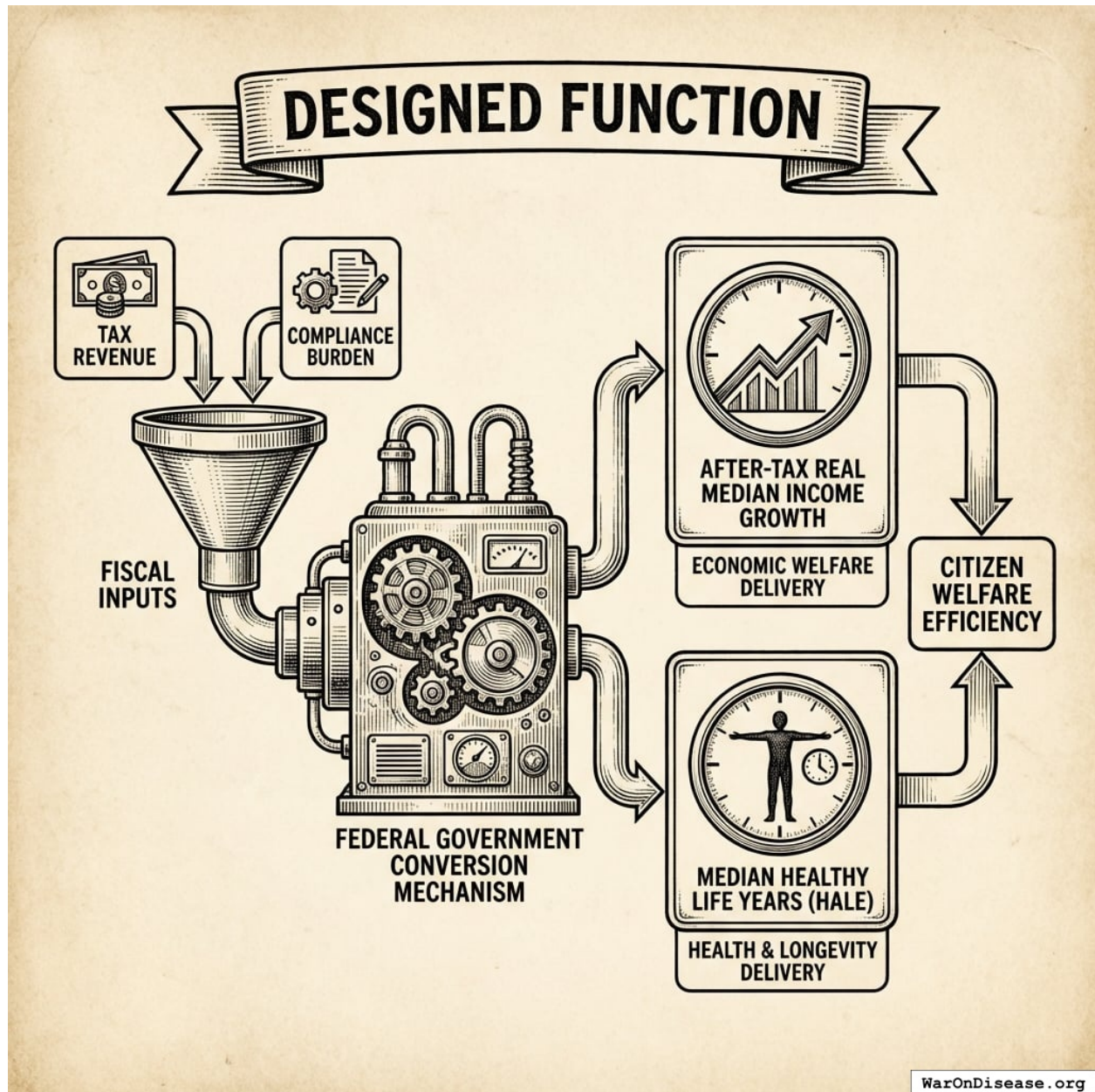


Figure 3: A flowchart illustrating the conversion process where fiscal inputs like tax revenue and compliance burdens are transformed by the federal government into terminal outcomes of median income growth and healthy life years.

The federal government's designed function is to convert fiscal inputs (tax revenue + compliance burden) into citizen welfare. We measure this conversion efficiency using two terminal outcomes:

1. **After-tax real median income growth:** measures economic welfare delivery
2. **Median healthy life years (HALE):** measures health and longevity delivery

2.2 Why Two Metrics Are Sufficient

These two outputs capture all upstream factors that matter:

Upstream Factor	Manifests in Income	Manifests in Healthy Life
Security/Safety	Crime costs, property loss	Violence, injury, chronic stress
Environment	Disaster costs, remediation	Respiratory disease, cancer
Freedom	Economic choice, mobility	Health decisions, reduced stress
Social Trust	Lower transaction costs	Mental health, social support
Education	Human capital, productivity	Health literacy
Infrastructure	Productivity, opportunity	Access to care, environmental health

These are not omissions. They are *upstream variables* that manifest in terminal outcomes. Measuring income + health implicitly captures everything that affects citizen welfare.

WHY TWO METRICS ARE SUFFICIENT: Upstream Factors Manifesting in Terminal Outcomes

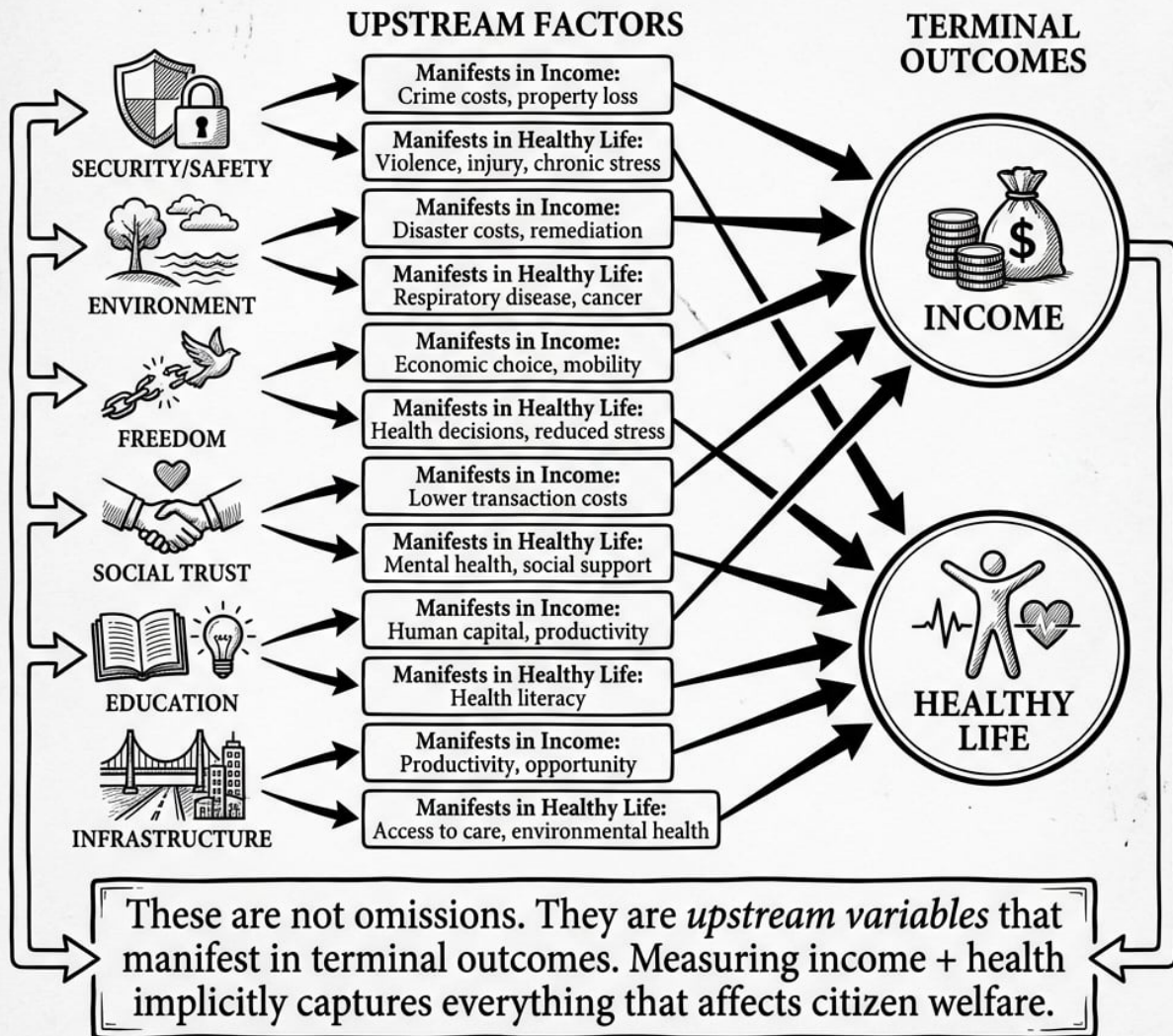


Figure 4: A conceptual diagram showing how multiple upstream factors like security, environment, and education funnel into two terminal outcomes: Income and Healthy Life.

2.3 Input-Output Measurement

Total System Input: ~\$6.75 trillion annually

- Federal revenue: \$4.9T¹³⁸
- State/local revenue attributable to federal mandates: ~\$1.3T¹³⁹
- Compliance burden (tax, regulatory): ~\$550B^{16,140}

Efficiency Metric: Output per dollar of input, benchmarked against OECD peer nations with comparable development levels.

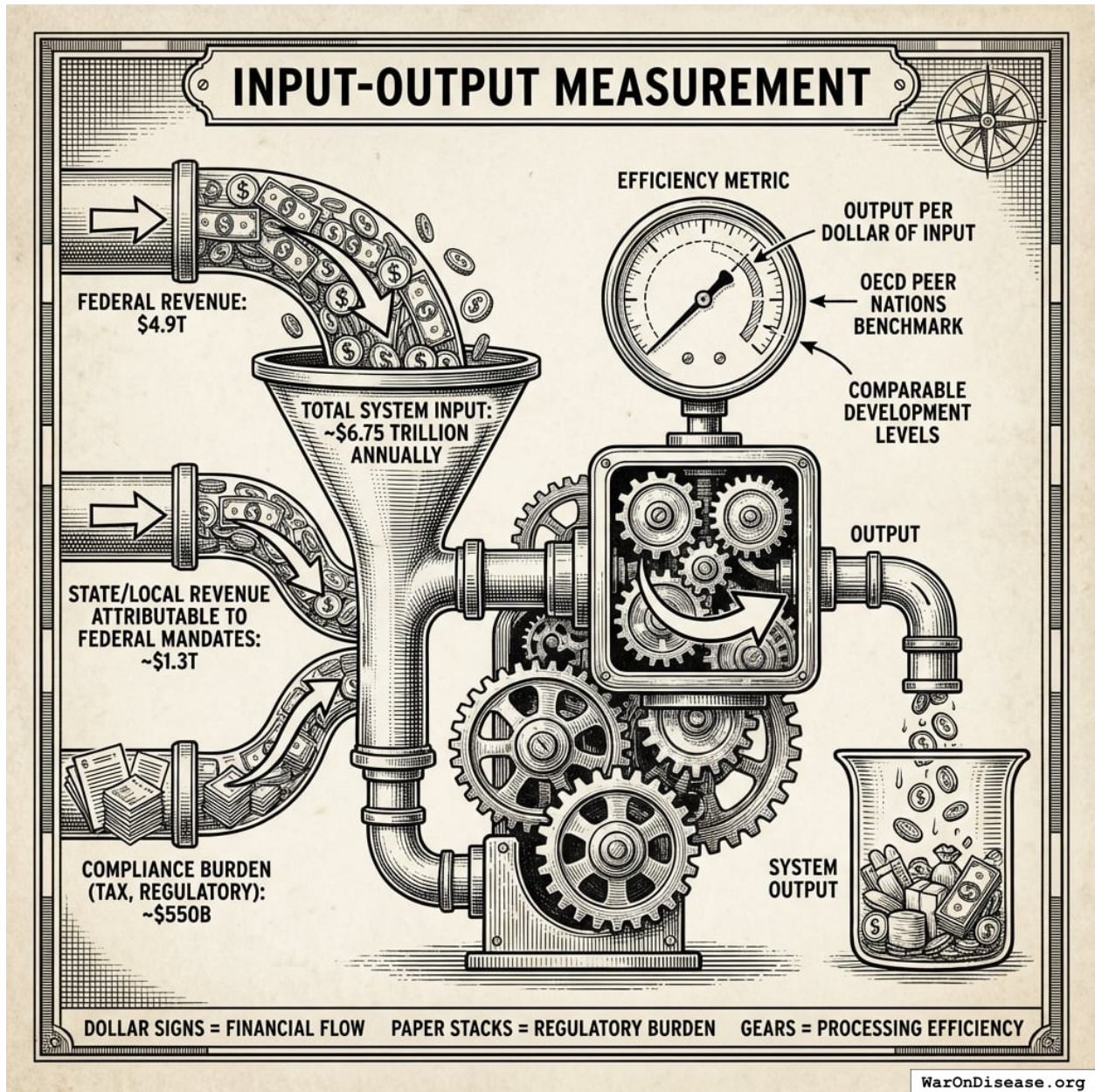


Figure 5: A breakdown of the \$6.75 trillion total system input, showing the proportional distribution of federal revenue, state and local mandates, and regulatory compliance burdens.

3 Methodology

3.1 Engineering Loss Categories

We categorize resource losses using engineering terminology rather than political language:

Loss Category	Definition	Examples
Friction Losses	Administrative overhead exceeding minimum necessary	Healthcare billing complexity, tax compliance burden
Leakage	Fraud, improper payments, unverified expenditure	Medicare improper payments, unaudited DoD assets
Parasitic Load	Bureaucracy maintaining itself rather than serving function	Redundant agencies, regulatory capture
Transmission Loss	Efficiency loss in federal → state → local → citizen transfer	Grant administration overhead, unfunded mandates
Idle/Standby Loss	Capacity maintained but unused	Excess military bases, redundant weapons systems
Conversion Inefficiency	Policy intent failing to achieve stated outcome	Drug interdiction not reducing use
Negative Work	Policies producing net harm rather than benefit	Incarceration increasing recidivism

3.2 Aggregate Efficiency Gap Calculation

The Aggregate Efficiency Gap (AEG) sums losses across all categories:

$$\text{AEG} = \sum_i \text{Friction}_i + \sum_j \text{Leakage}_j + \sum_k \text{Parasitic}_k + \sum_l \text{Transmission}_l + \sum_m \text{Idle}_m + \sum_n \text{Conversion}_n + \sum_o \text{Negative}_o$$

We employ Monte Carlo simulation to generate confidence intervals, recognizing uncertainty in loss estimates (particularly where data opacity exists, such as the DoD’s inability to audit 61% of assets).

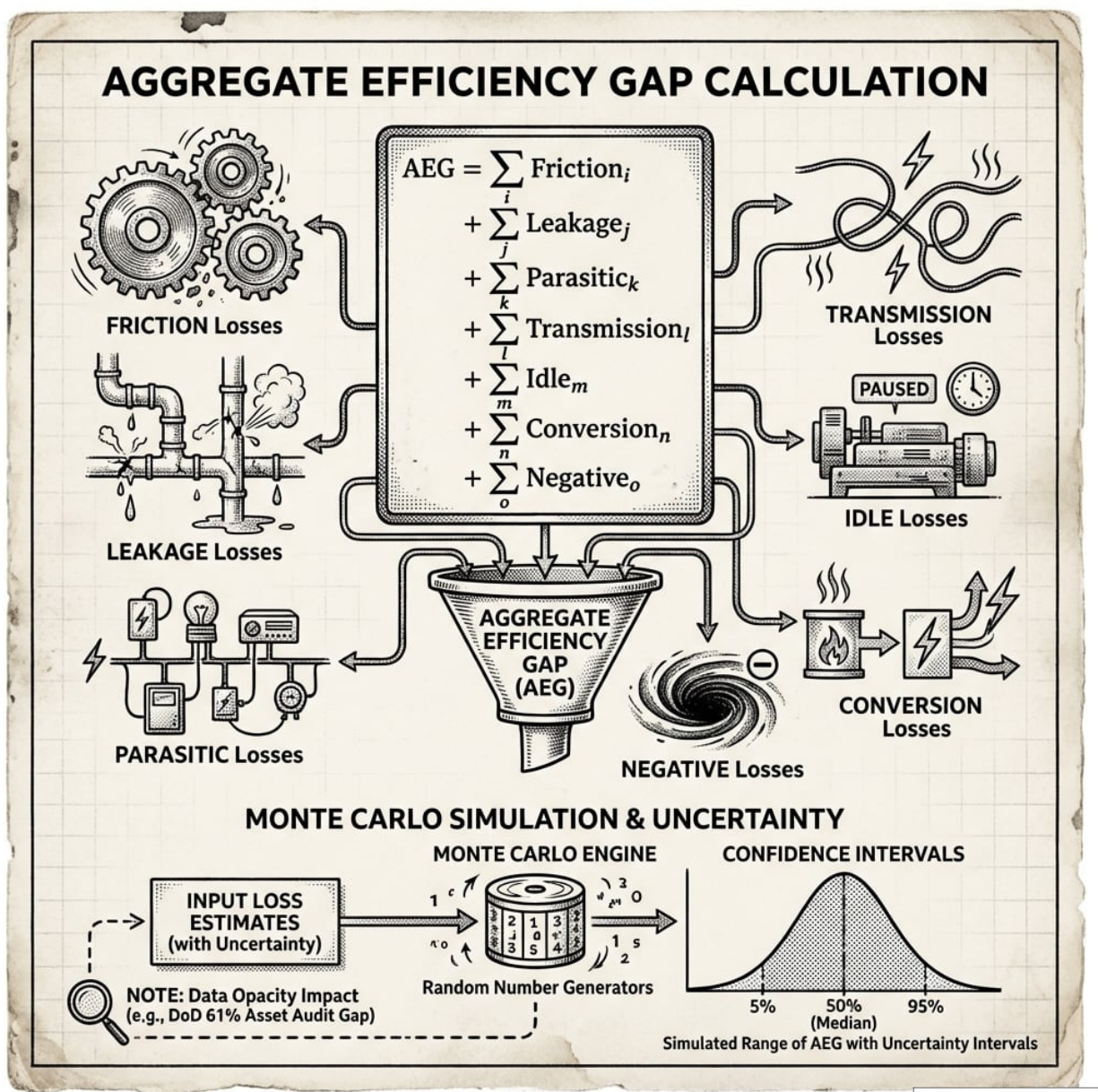
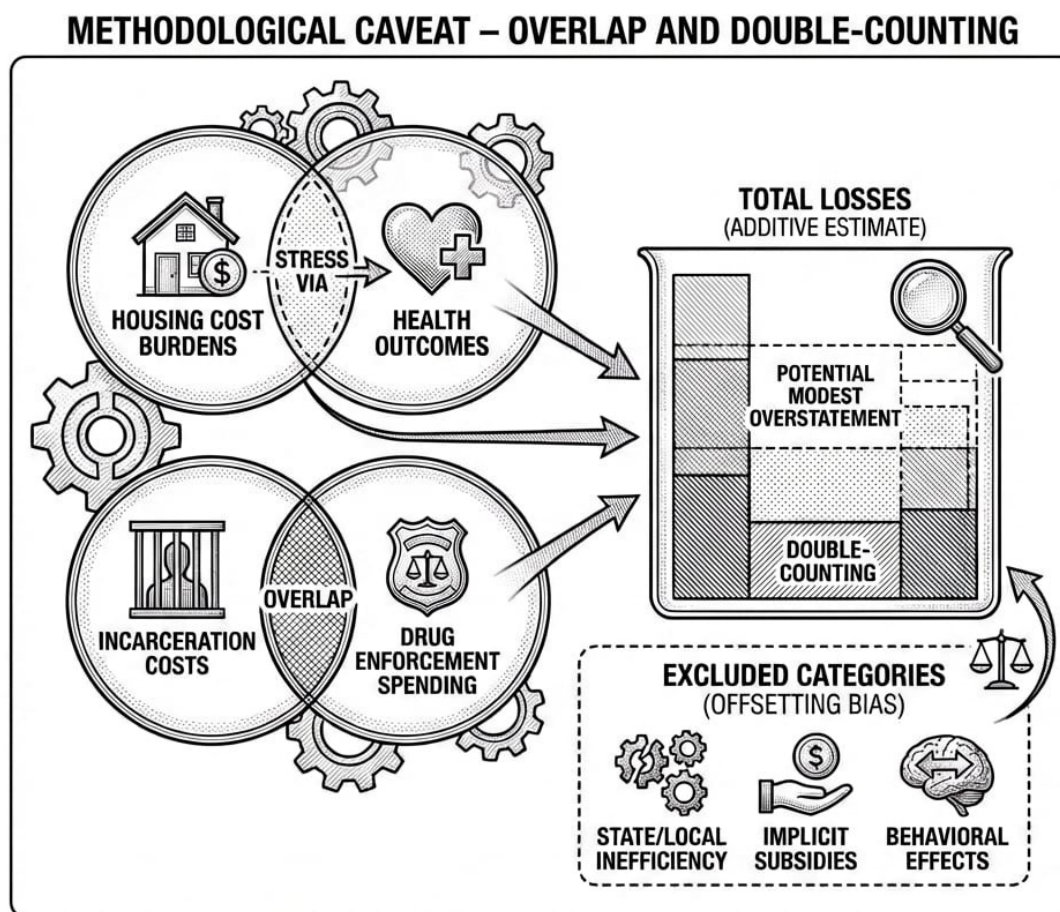


Figure 6: A structural diagram showing the seven categories of efficiency loss aggregating into the AEG, with visual indicators for the high uncertainty (61%) addressed by Monte Carlo simulation.



Methodological Caveat: Overlap and Double-Counting. While subsystem losses are estimated independently, some interdependence exists. For example, housing cost burdens affect health outcomes via stress; incarceration costs overlap with drug enforcement spending. This analysis treats categories as largely additive, which may modestly overstate total losses. However, excluded categories (state/local inefficiency, implicit subsidies, behavioral effects) likely offset this bias.

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Figure 7: A conceptual diagram illustrating the overlaps between subsystem costs like housing and health, contrasted with excluded factors that offset the resulting bias.

While subsystem losses are estimated independently, some interdependence exists. For example, housing cost burdens affect health outcomes via stress; incarceration costs overlap with drug enforcement spending. This analysis treats categories as largely additive, which may modestly overstate total losses. However, excluded categories (state/local inefficiency, implicit subsidies, behavioral effects) likely offset this bias.

3.3 Valuation Standards

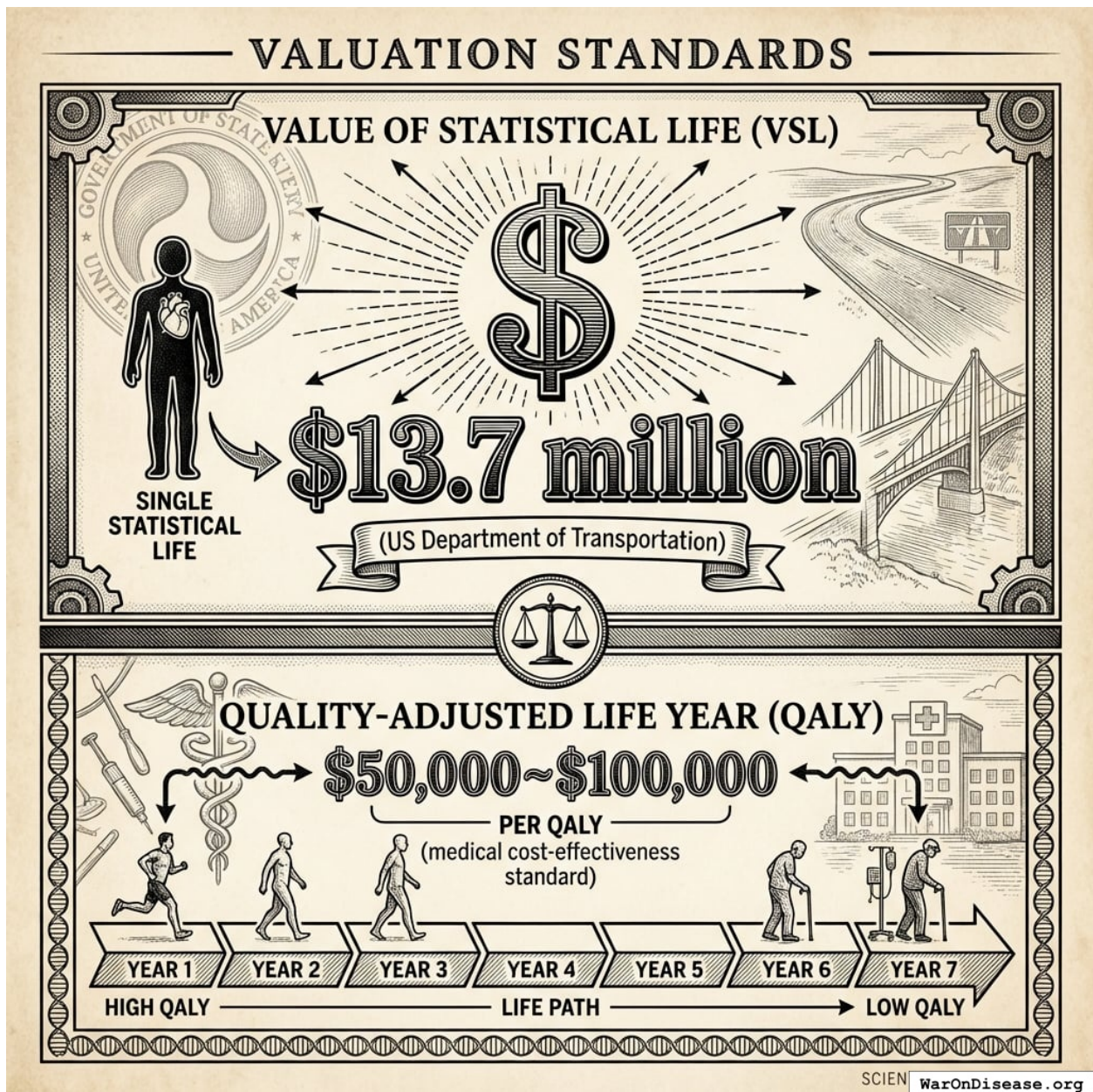


Figure 8: A comparison showing the massive scale difference between the \$13.7 million Value of Statistical Life (VSL) and the \$50,000–\$100,000 Quality-Adjusted Life Year (QALY) standards.

- **Value of Statistical Life (VSL):** \$13.7 million (US Department of Transportation)¹⁴¹
- **Quality-Adjusted Life Year (QALY):** \$50,000–\$100,000 (medical cost-effectiveness standard)¹⁴²

4 Subsystem Audit: Defense

The Department of Defense operates as the largest discretionary expenditure node, with annual spending of approximately \$900 billion¹⁴³. Current spending exceeds the next nine nations combined¹⁴⁴.

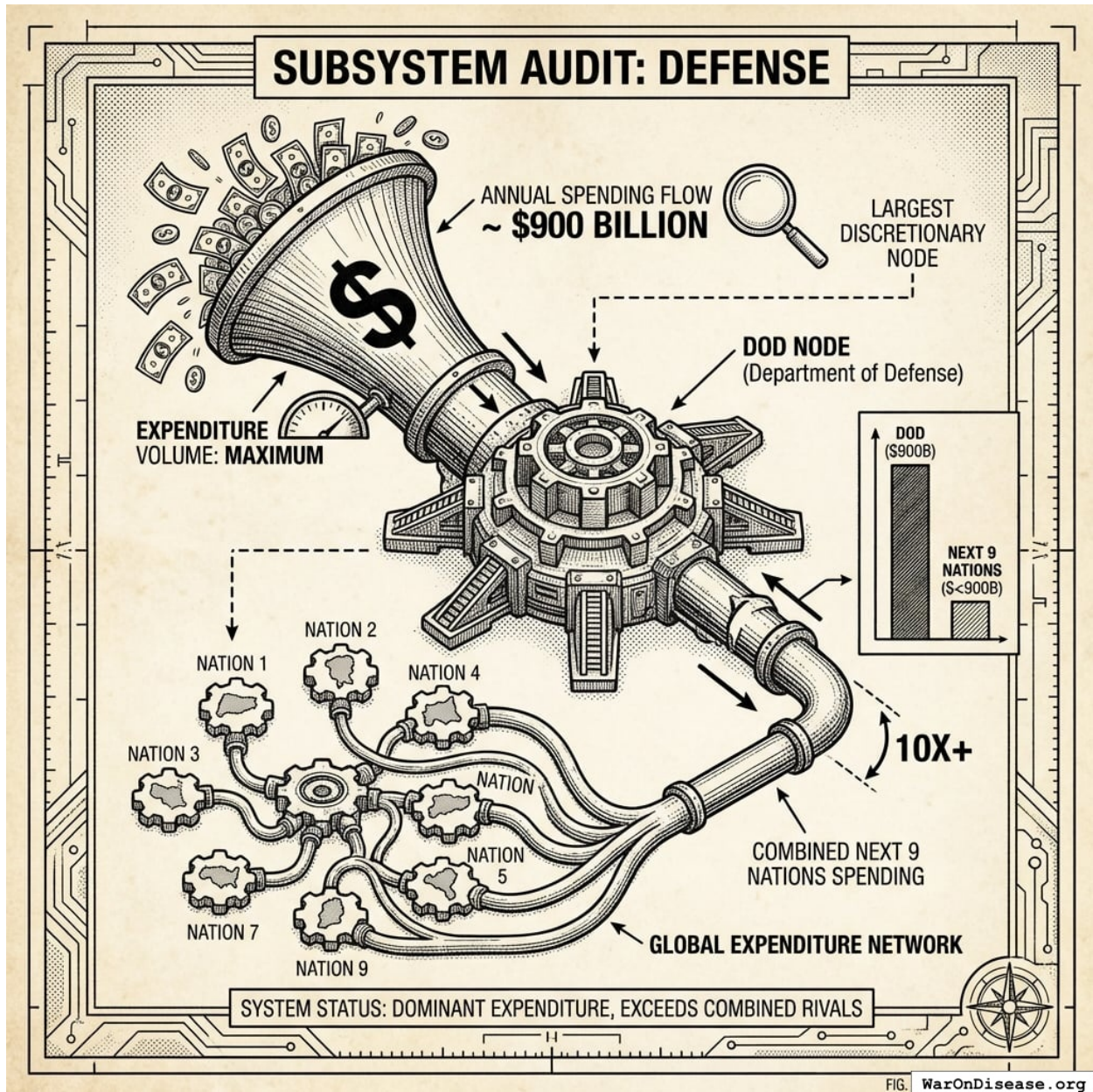


Figure 9: A comparison of the \$900 billion U.S. defense budget against the combined military expenditures of the next nine highest-spending nations.

4.1 Loss Category: Leakage (Audit Failure)

In November 2024, the Pentagon failed its seventh consecutive audit¹⁴⁵. The DoD was unable to account for 61-63% of its \$3.8 trillion in assets¹⁴⁶, approximately \$2.5 trillion in property, equipment,

and inventory with unknown status.

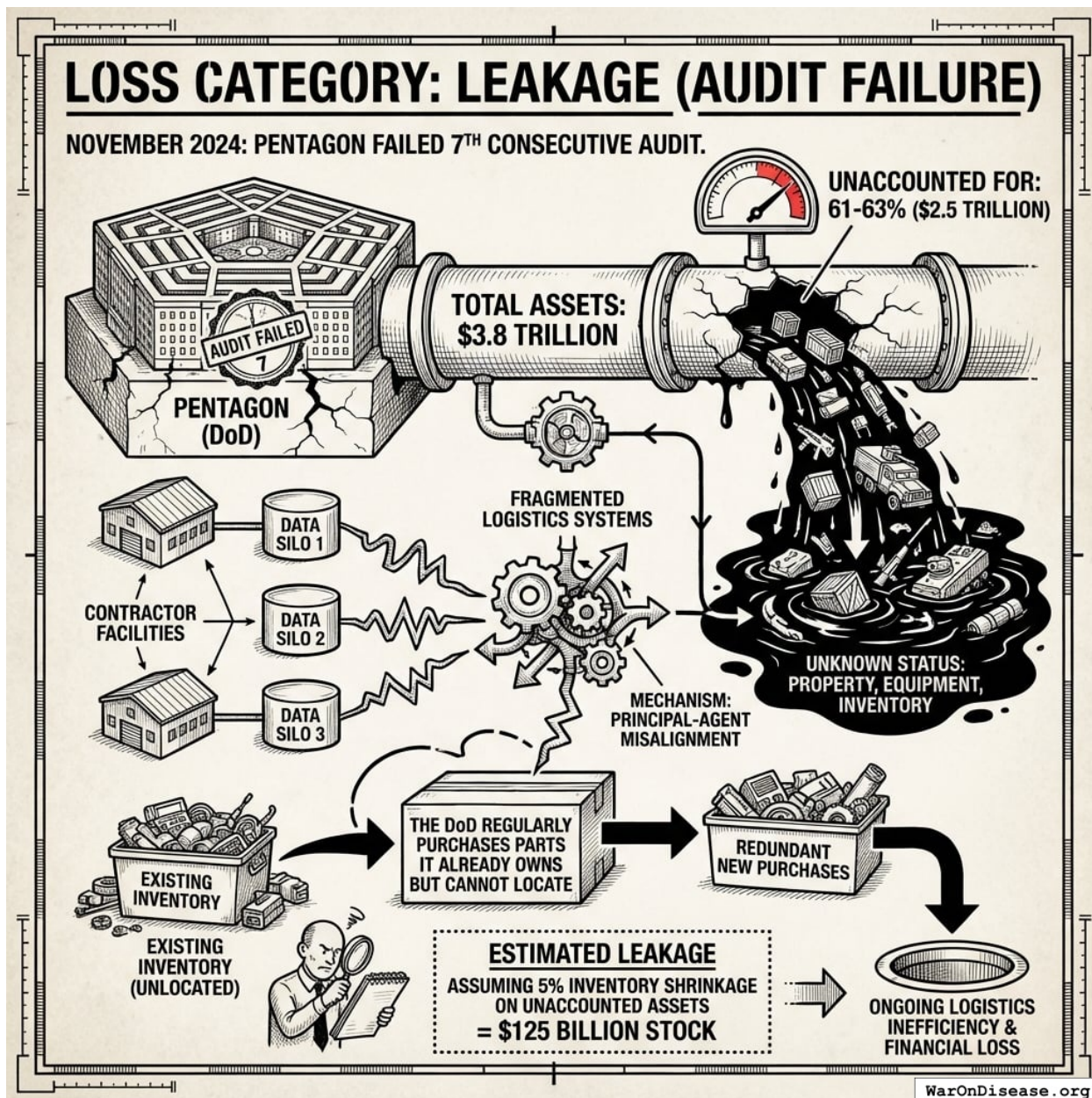


Figure 10: A breakdown of the Department of Defense's \$3.8 trillion asset pool showing the 61-63% (\$2.5 trillion) portion that is currently unaccounted for, leading to an estimated \$125 billion in inventory shrinkage.

The mechanism: fragmented logistics systems where contractors record inventory data, creating principal-agent misalignment¹⁴⁷. Without verified asset ledgers, the DoD regularly purchases parts it already owns but cannot locate.

Estimated leakage: Assuming 5% inventory shrinkage on unaccounted assets = \$125 billion stock, plus ongoing logistics inefficiency.

4.2 Loss Category: Conversion Inefficiency (F-35 Program)

The F-35 program exemplifies “concurrency”: producing aircraft before design completion. Results:

- Lifetime sustainment cost: increased from \$1.1T (2018) to \$1.58T (2023)¹⁴⁸
- Availability rates: declining despite 44% cost increase¹⁴⁸
- 2024 delivery delays: average 238 days late¹⁴⁹

Estimated conversion inefficiency: \$15-20 billion annually in defect remediation and unflown flight hours.

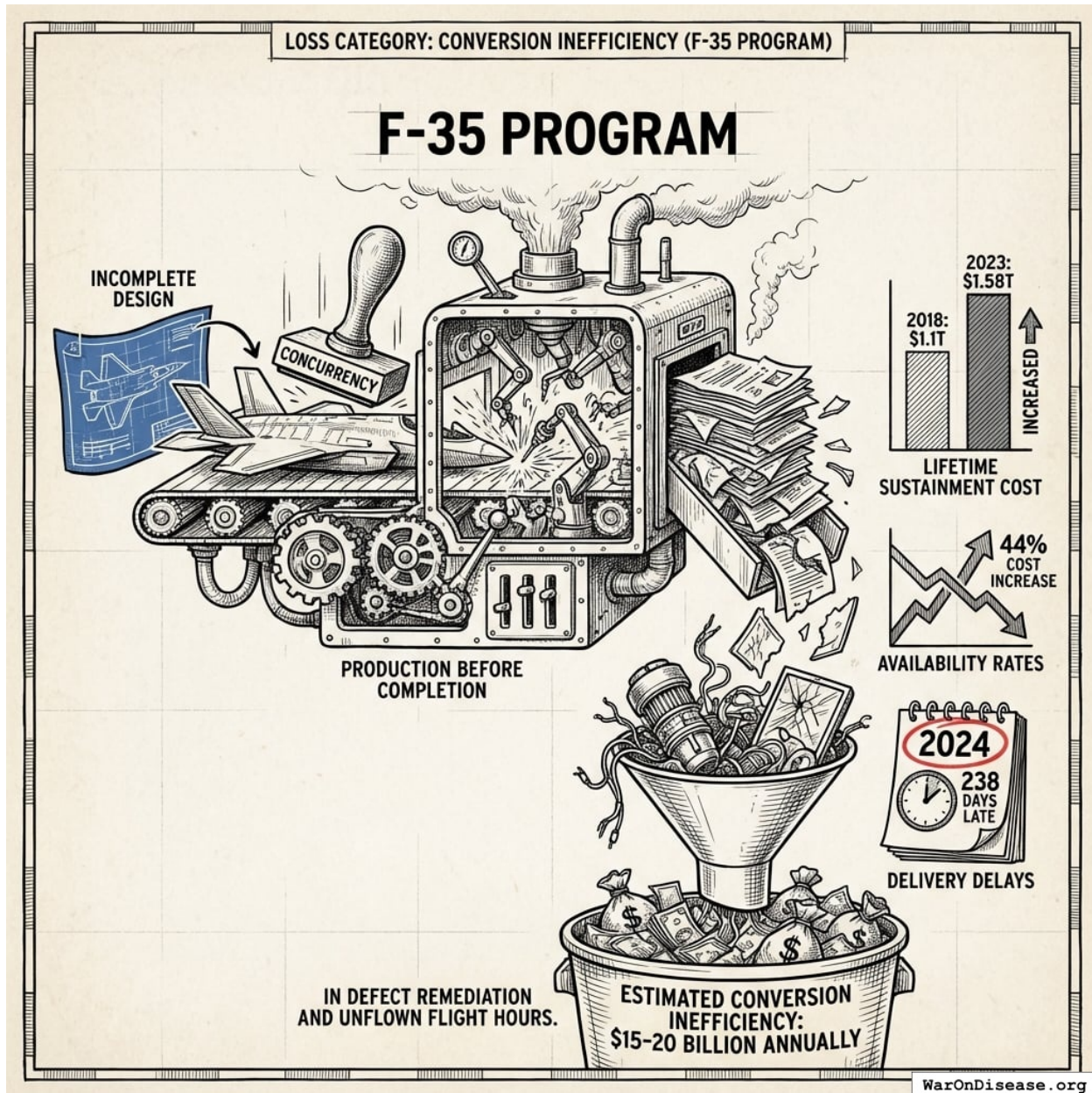


Figure 11: A comparison of rising F-35 lifetime sustainment costs against declining availability rates, highlighting the \$15-20 billion annual loss from conversion inefficiency.

4.3 Loss Category: Idle/Standby (Overseas Basing)

The U.S. maintains approximately 750 military bases in over 80 countries¹⁵⁰, architecture designed for 1945 geopolitics.

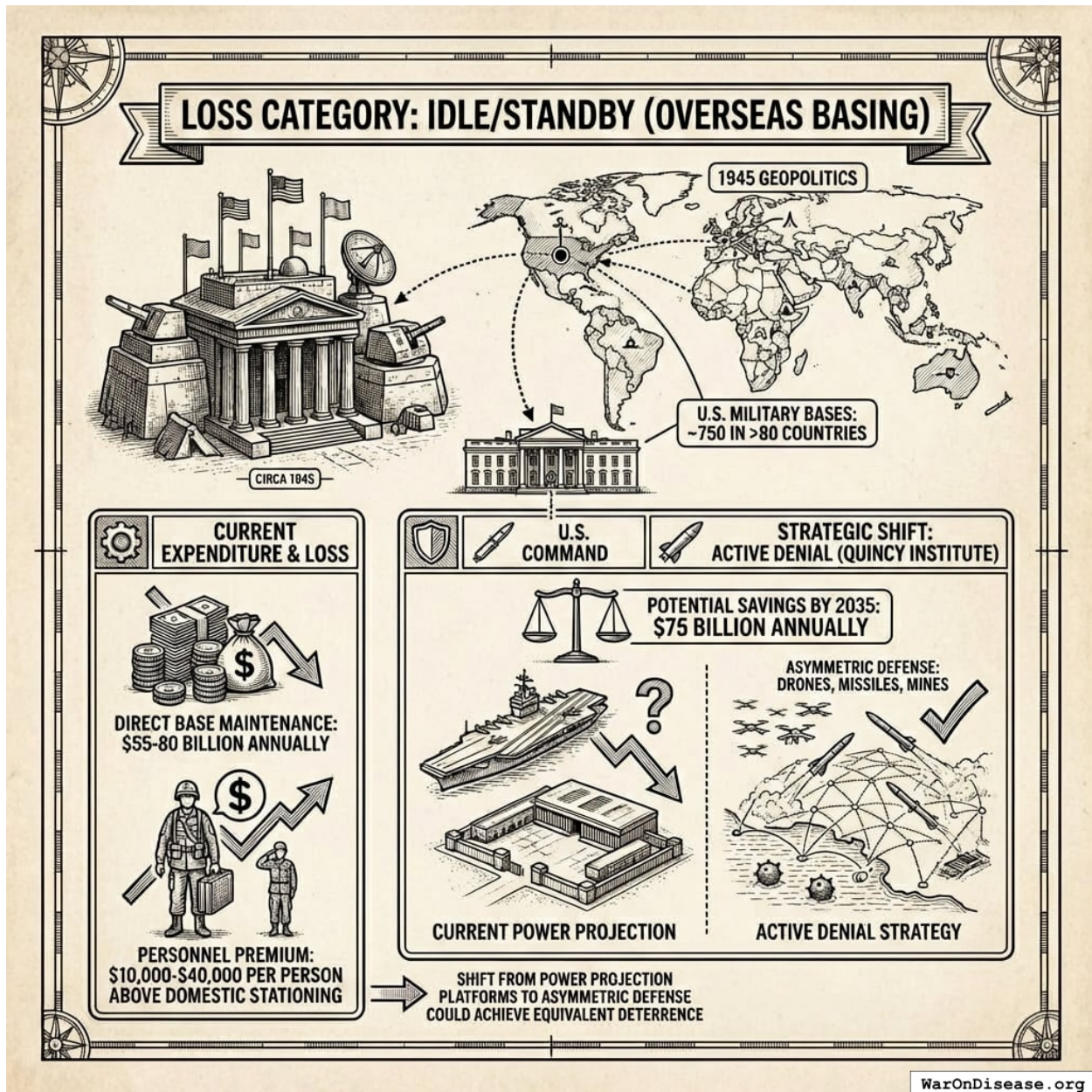


Figure 12: A bar chart comparing the current \$55-80 billion annual cost of overseas base maintenance against the projected \$75 billion in potential annual savings from transitioning to an Active Denial strategy.

- Direct base maintenance: \$55-80 billion annually¹⁵⁰
- Personnel premium: \$10,000-\$40,000 per person above domestic stationing¹⁵¹

The Quincy Institute estimates that shifting to “Active Denial” strategy (asymmetric defense via drones, missiles, mines rather than power projection platforms) could achieve equivalent deterrence

at \$75 billion annual savings by 2035¹⁵².

4.4 Loss Category: Parasitic (Strategic Misalignment)

The “Overmatch” doctrine requires dominance in every theater simultaneously, creating unlimited spending requirements. The Congressional Budget Office projects defense costs will rise to \$965 billion by 2039¹⁵³, driven by this refusal to rationalize legacy commitments. A rational optimization would focus on:

- Robust nuclear deterrent (submarine-based leg sufficient)
- Naval denial capabilities
- Asymmetric defense posture

Cutting the redundant ICBM leg and reducing bomber procurement would save \$15-20 billion annually¹⁵⁴.

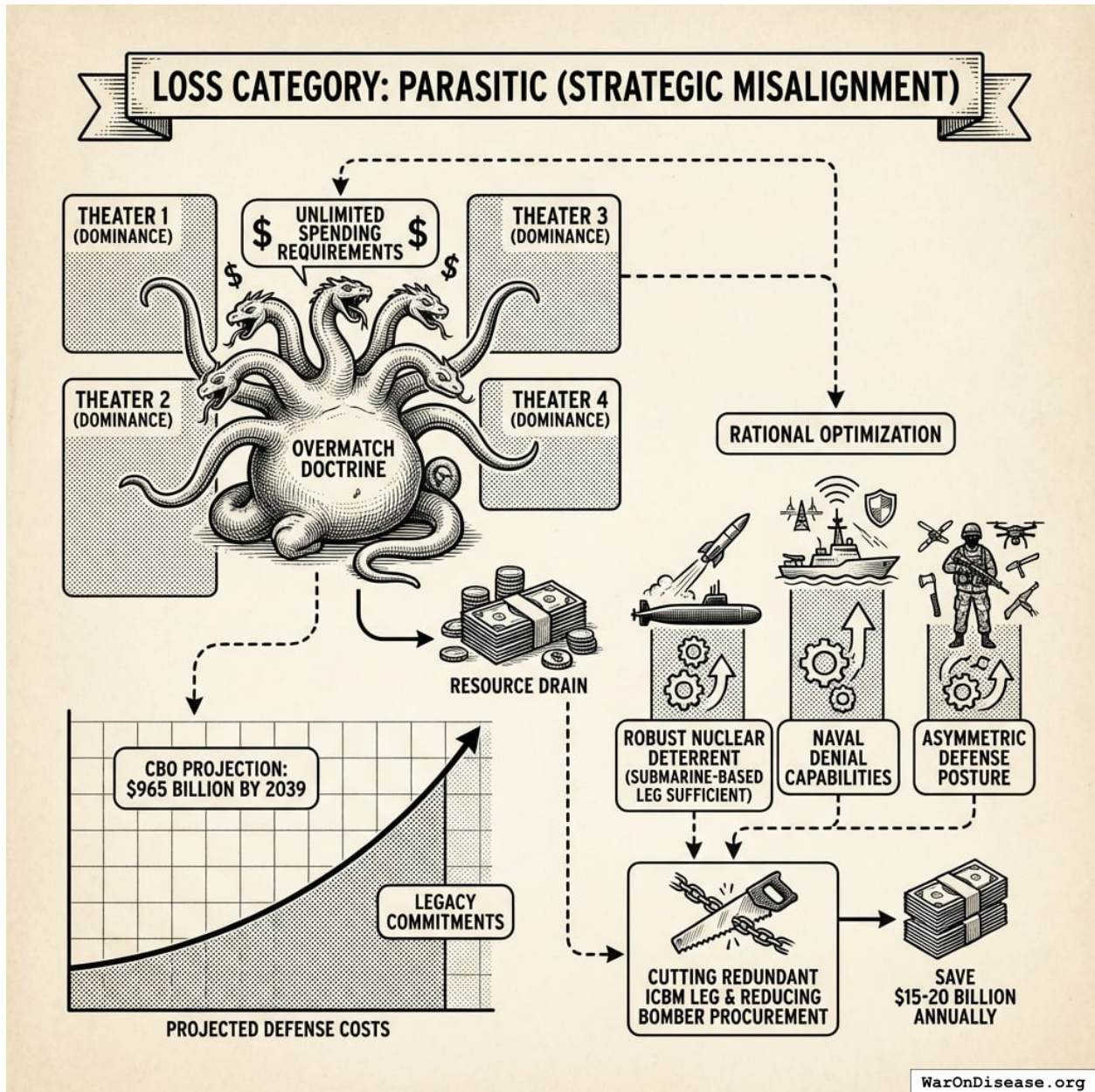


Figure 13: A comparison of the projected \$965 billion defense budget against the potential \$15-20 billion in annual savings from strategic nuclear cuts.

4.5 Defense Subsystem Summary

Loss Category	Low Estimate	Mean	High Estimate
Leakage (audit)	\$50B	\$75B	\$100B
Conversion (F-35)	\$15B	\$17B	\$20B
Idle (bases)	\$40B	\$55B	\$75B
Parasitic (strategy)	\$50B	\$63B	\$105B
Total	\$100B	\$210B	\$300B

5 Subsystem Audit: Healthcare Administration

U.S. healthcare consumes ~18% of GDP (\$5.3 trillion)¹⁵⁵ yet delivers health outcomes inferior to peer nations spending 10-11% of GDP¹⁵⁶. The delta is not care quality. It is administrative friction.

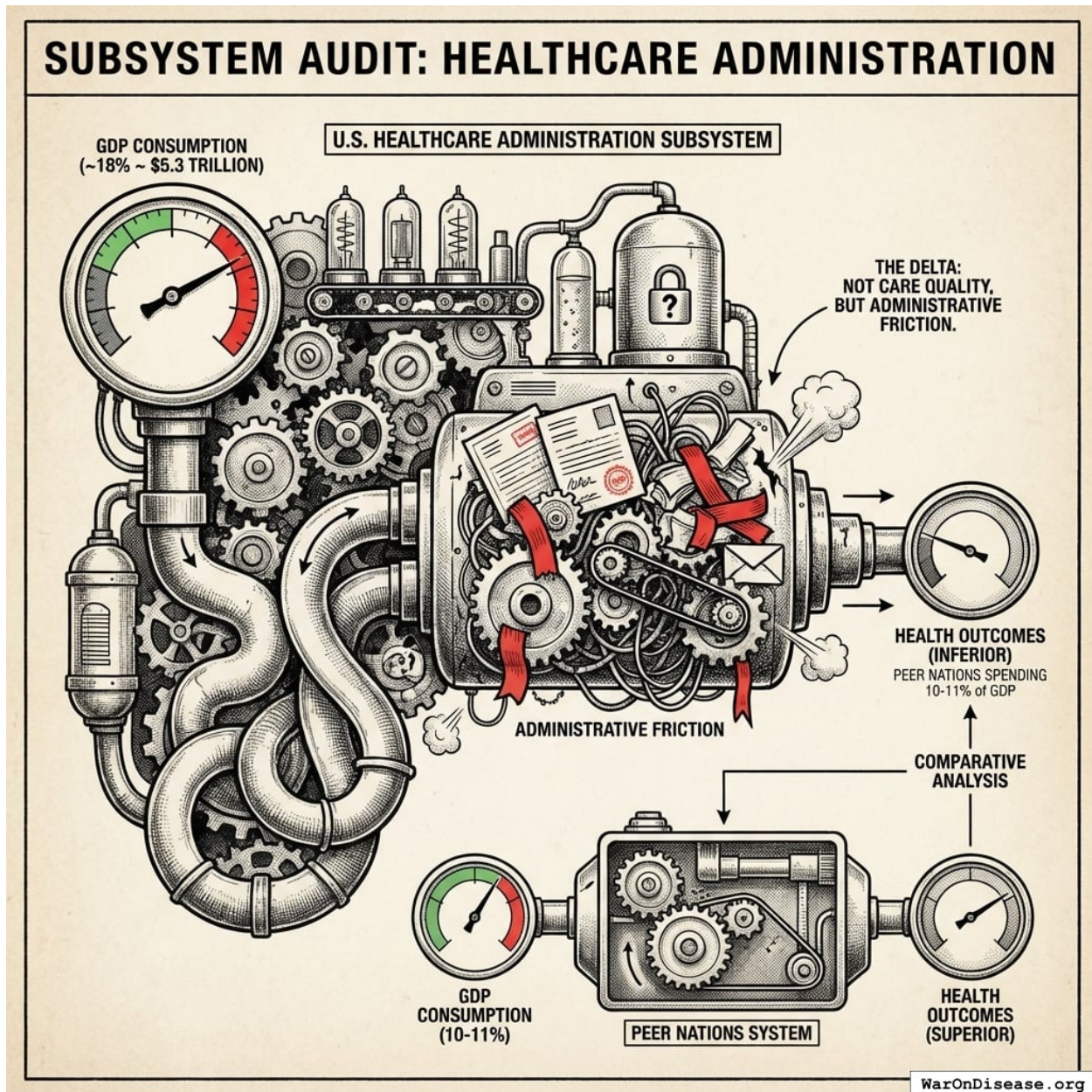


Figure 14: A comparison of healthcare spending as a percentage of GDP between the U.S. (18%) and peer nations (10-11%), highlighting the financial discrepancy linked to administrative overhead.

5.1 Loss Category: Friction (Administrative Overhead)

The U.S. spends approximately \$1,000 more per person on administrative costs than the average wealthy OECD country¹⁵⁶. With 335 million population:

Administrative excess: ~\$335 billion annually

A 2020 study found U.S. administrative spending at 34.2% of health expenditures versus 17% in Canada¹⁵⁷. This overhead does not improve outcomes. It diverts resources from care to paperwork.

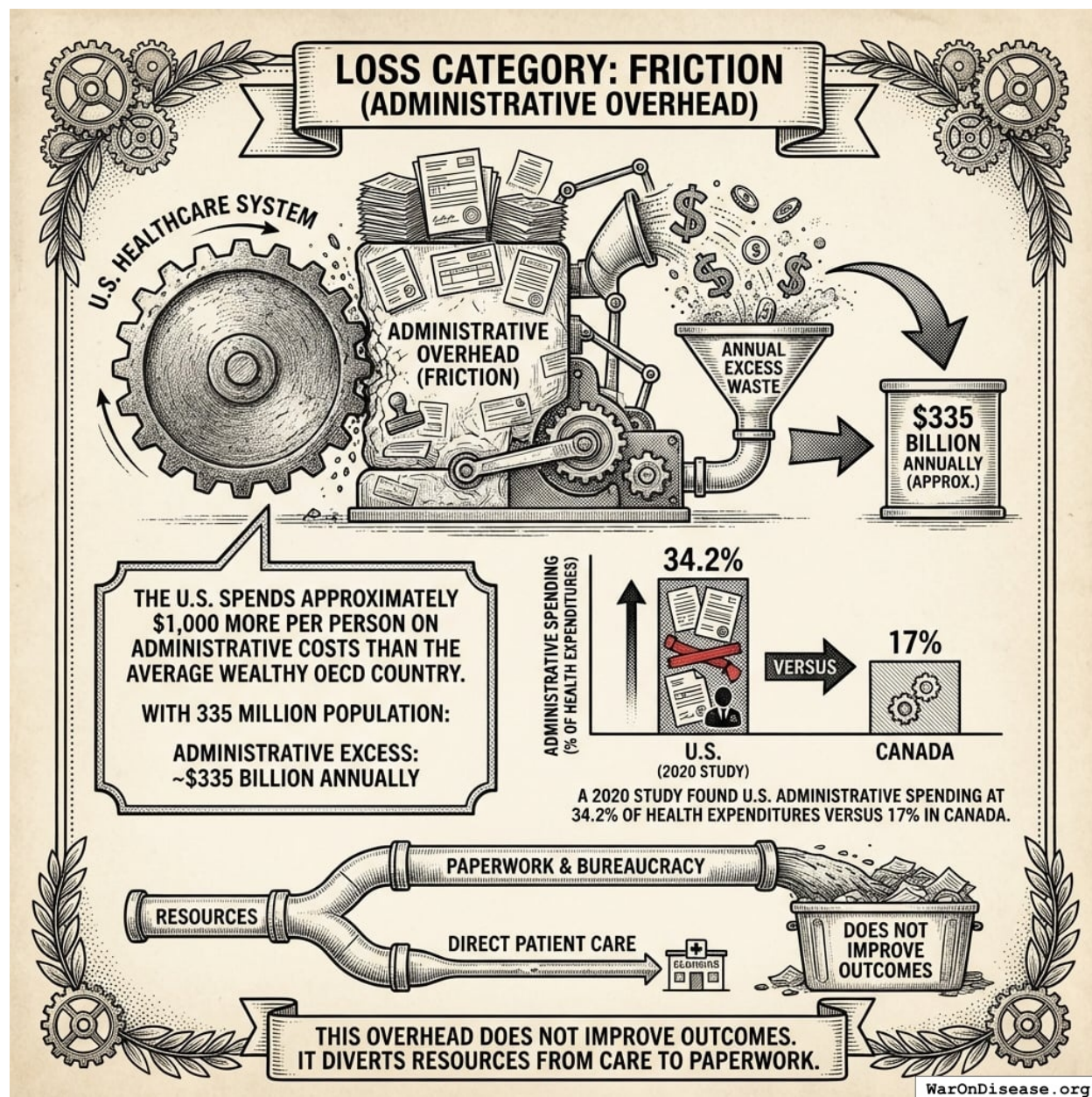


Figure 15: Comparison of healthcare administrative costs showing the U.S. spending 34.2% of expenditures on overhead versus 17% in Canada, alongside the resulting \$335 billion in annual excess spending.

5.2 Loss Category: Leakage (Medicare Advantage Upcoding)

Medicare Advantage functions as a subsidy mechanism via “upcoding”: making patients appear sicker than they are to increase capitated payments.

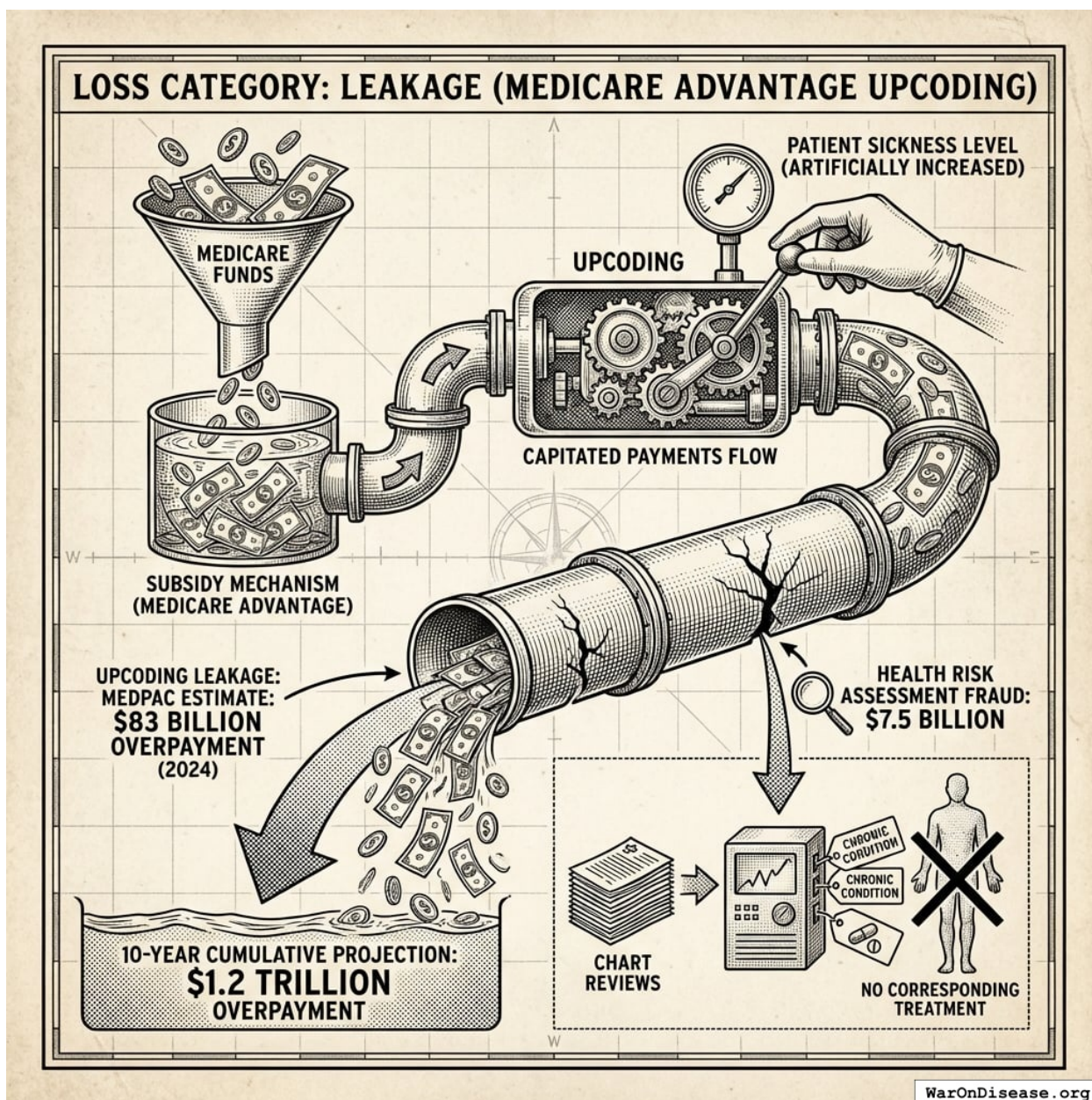


Figure 16: A comparison of estimated Medicare Advantage overpayments, highlighting the \$83 billion annual cost for 2024 against a \$1.2 trillion ten-year projection and \$7.5 billion in specific fraud categories.

- MedPAC estimate: \$83 billion overpayment in 2024¹⁵⁸
- 10-year projection: \$1.2 trillion cumulative overpayment¹⁵⁹
- Health Risk Assessment fraud: \$7.5 billion from chart reviews generating diagnoses with no corresponding treatment¹⁶⁰

5.3 Loss Category: Leakage (Improper Payments)

GAO estimates \$162 billion in improper payments government-wide in 2024, with 75% concentrated in Medicare and Medicaid¹⁶¹.

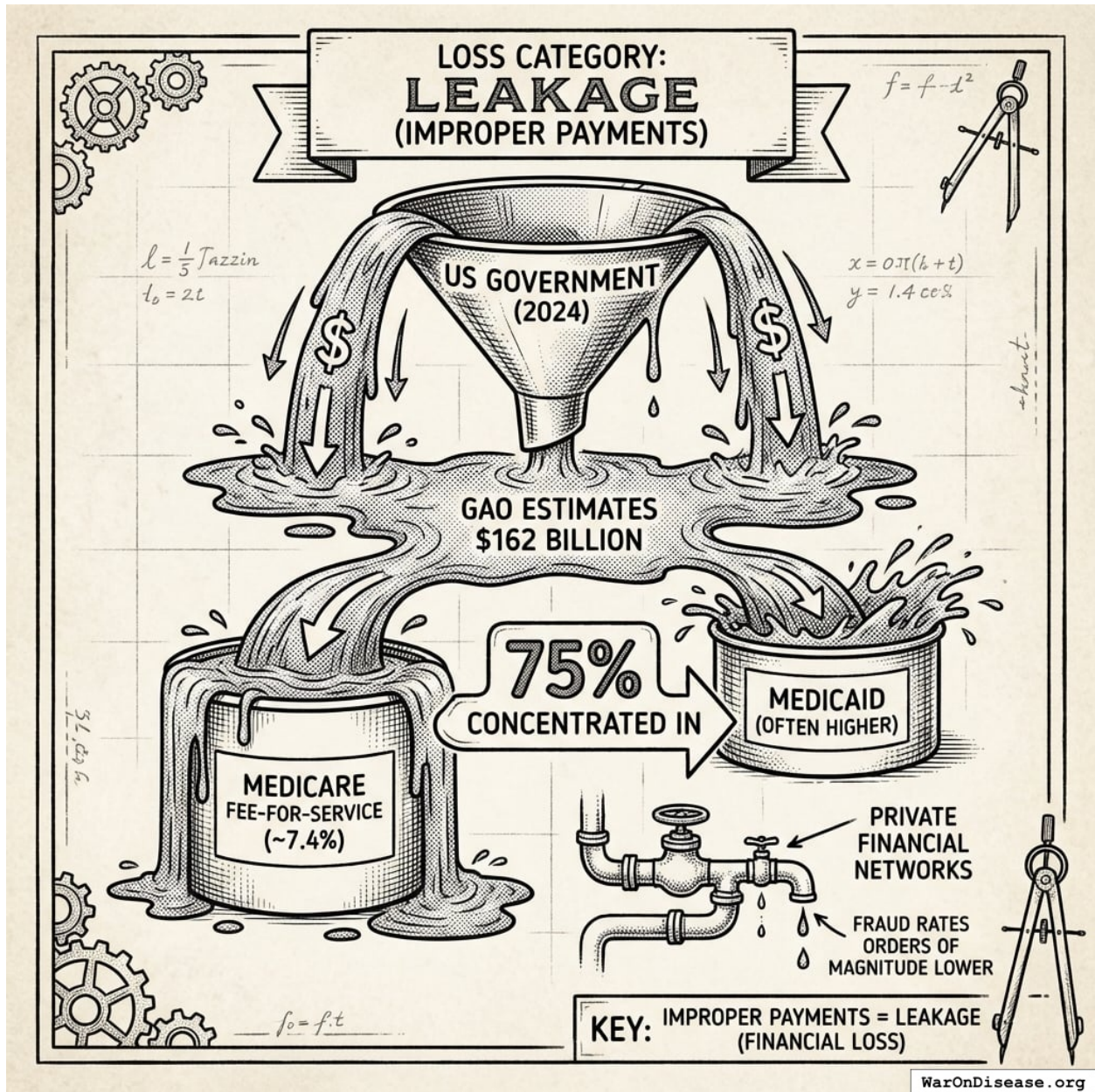


Figure 17: A comparison of government improper payment rates, specifically highlighting the 75% concentration in Medicare and Medicaid against significantly lower private sector fraud benchmarks.

- Medicare Fee-for-Service: ~7.4% improper payment rate
- Medicaid: often higher

Private financial networks operate with fraud rates orders of magnitude lower.

5.4 Healthcare Subsystem Summary

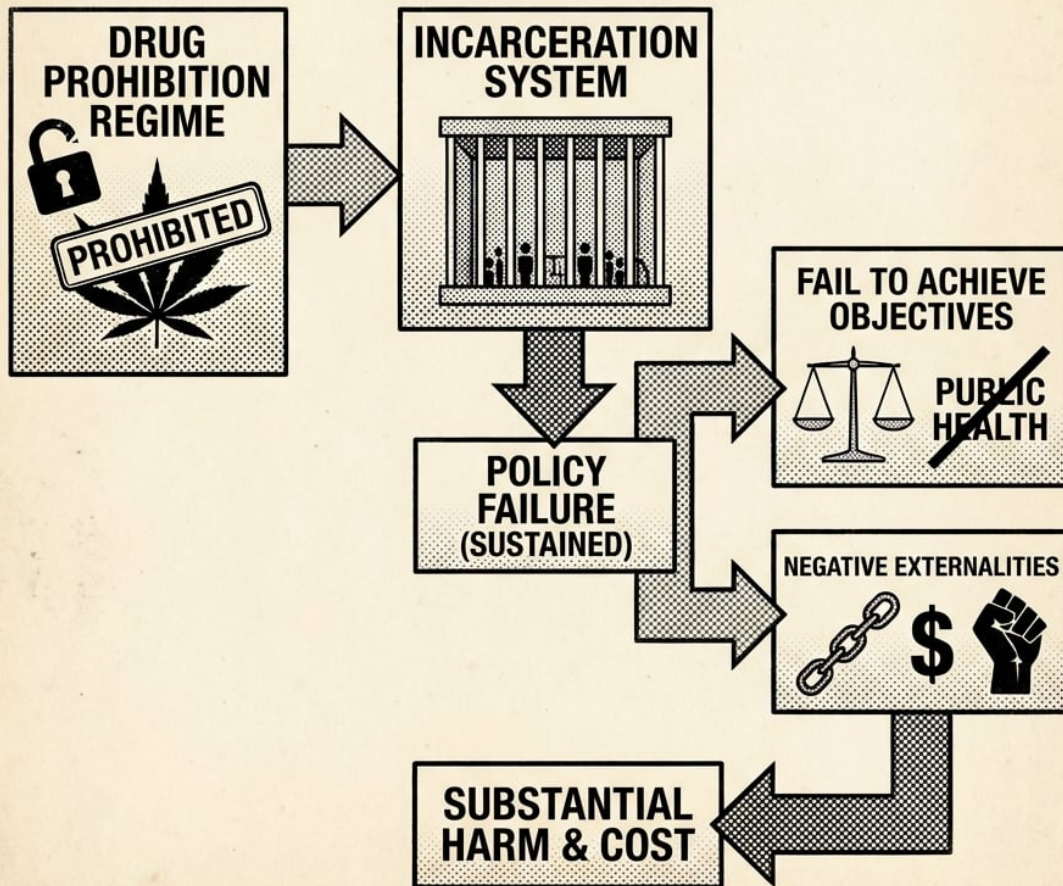
Loss Category	Low Estimate	Mean	High Estimate
Friction (admin)	\$250B	\$335B	\$400B

Loss Category	Low Estimate	Mean	High Estimate
Leakage (MA upcoding)	\$60B	\$83B	\$100B
Leakage (improper payments)	\$80B	\$120B	\$150B
Total	\$300B	\$450B	\$600B

6 Subsystem Audit: Justice and Prohibition

The drug prohibition regime and resulting incarceration system represent sustained policy failure: intervention that fails to achieve stated objectives while generating substantial negative externalities.

SUBSYSTEM AUDIT: JUSTICE AND PROHIBITION



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Figure 18: A conceptual diagram illustrating the failure of the drug prohibition system, showing the gap between intended policy objectives and the actual outcomes of incarceration and negative externalities.

6.1 Loss Category: Conversion Inefficiency (Drug Prohibition)

The federal drug control budget for 2024: nearly \$45 billion¹⁶². Total expenditure since 1971: over \$1 trillion¹⁶³.

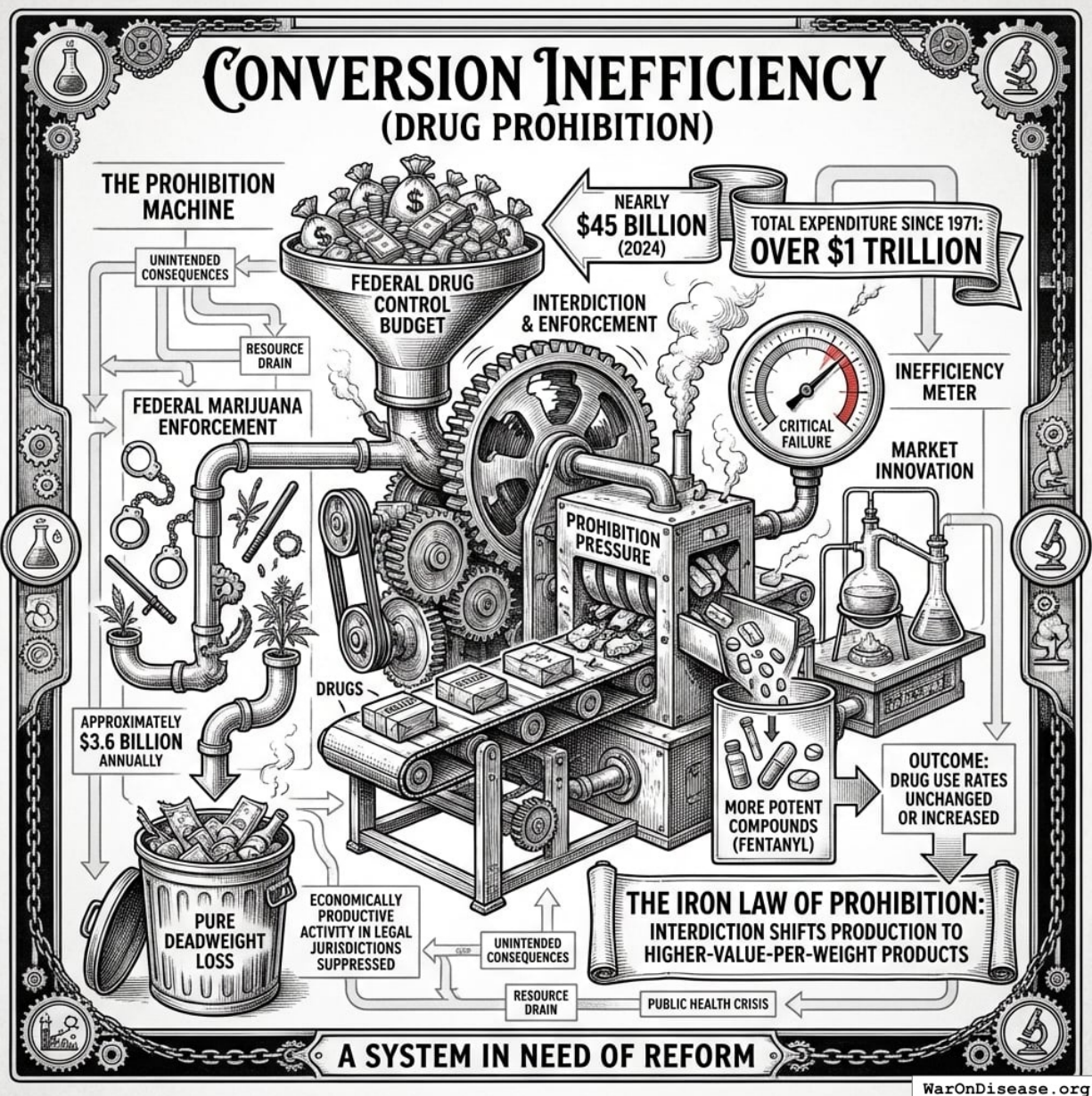


Figure 19: A comparison of massive federal drug control spending (\$1 trillion cumulative, \$45 billion annual) against stagnant usage rates and the shift toward higher-potency drugs like fentanyl under the Iron Law of Prohibition.

Outcome: Drug use rates unchanged or increased¹⁶⁴. The market has innovated toward more potent compounds (fentanyl). This is a direct consequence of the “Iron Law of Prohibition,” where interdiction shifts production to higher-value-per-weight products.

Despite state legalization, federal marijuana enforcement continues at approximately \$3.6 billion annually¹⁶⁵, pure deadweight loss on activity that is economically productive in legal jurisdictions.

6.2 Loss Category: Negative Work (Mass Incarceration)

The U.S. incarcerates at rates unmatched in the developed world, removing prime-age workers from the labor force and degrading human capital.

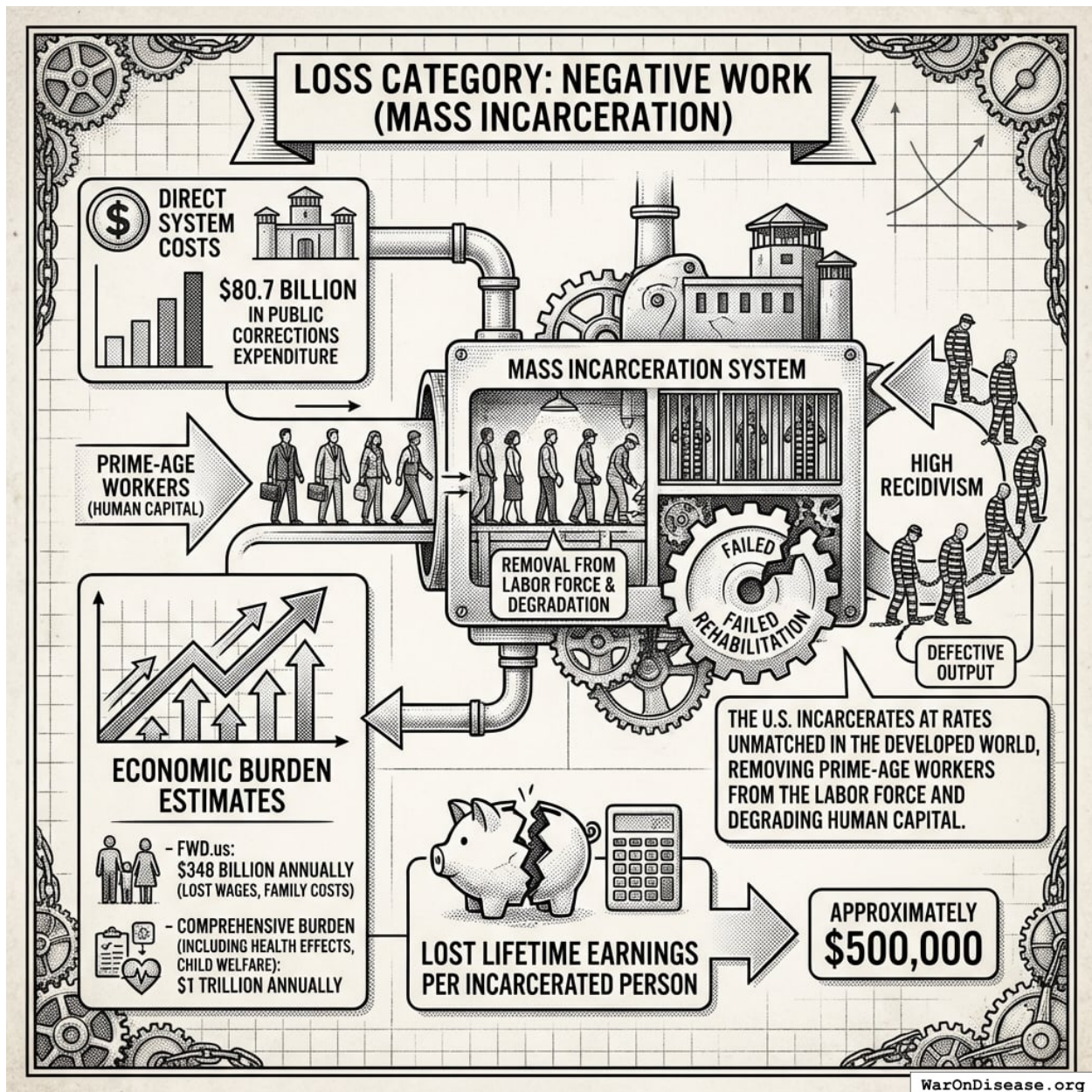


Figure 20: A comparison of the tiered economic impacts of mass incarceration, contrasting direct public corrections costs against broader societal burdens and lost lifetime earnings.

Direct system costs: \$80.7 billion in public corrections expenditure¹⁶⁶

Economic burden estimates:

- FWD.us: \$348 billion annually (lost wages, family costs)¹⁶⁷

- Comprehensive burden (including health effects, child welfare): \$1 trillion annually¹⁶⁸

Lost lifetime earnings per incarcerated person: approximately \$500,000¹⁶⁶. The system fails at rehabilitation. High recidivism rates mean the “correctional” investment yields defective output.

6.3 Loss Category: Leakage (Civil Asset Forfeiture)

Civil asset forfeiture allows property seizure without criminal conviction. FY 2024 Treasury Forfeiture Fund: \$2.26 billion processed¹⁶⁹.

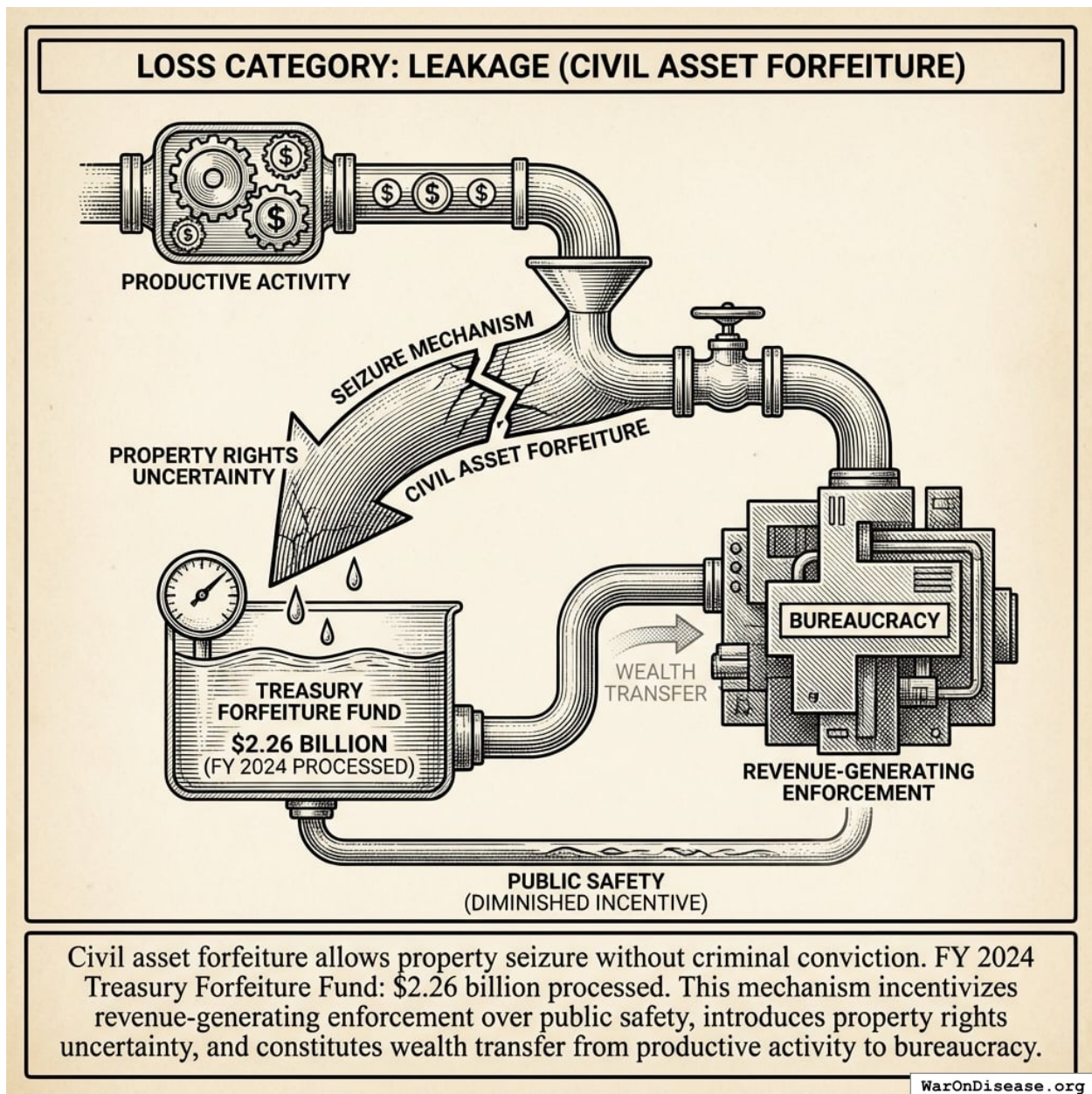


Figure 21: A conceptual flow diagram illustrating how civil asset forfeiture redirects property from productive economic activity into a bureaucratic fund, shifting enforcement incentives toward revenue generation.

This mechanism incentivizes revenue-generating enforcement over public safety, introduces property rights uncertainty, and constitutes wealth transfer from productive activity to bureaucracy.

6.4 Justice Subsystem Summary

Loss Category	Low Estimate	Mean	High Estimate
Conversion (drug enforcement)	\$40B	\$48B	\$55B
Negative work (incarceration)	\$80B	\$280B	\$500B
Leakage (forfeiture)	\$2B	\$2B	\$3B
Total	\$100B	\$280B	\$500B

Note: The 6x range (\$80B-\$500B) for incarceration reflects genuine uncertainty. The low bound captures only direct corrections spending; the high bound includes comprehensive economic burden estimates (lost wages, family disruption, intergenerational effects). Different methodologies yield dramatically different figures.

7 Subsystem Audit: Regulatory and Tax Compliance

The compliance burden (time and resources consumed meeting federal requirements) represents a substantial unrecorded subtraction from national output.

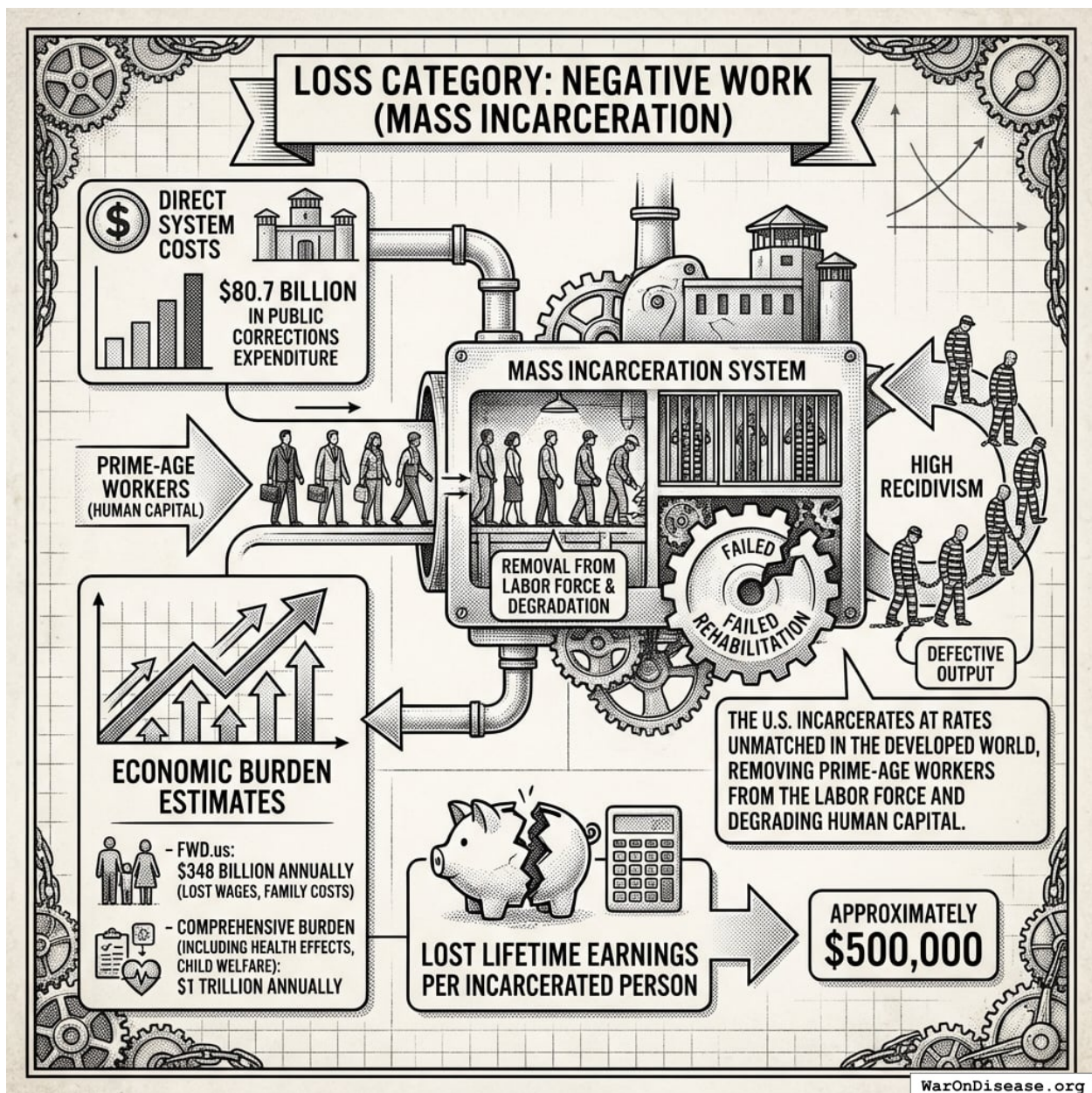


Figure 22: A comparison of the escalating economic impacts of mass incarceration, contrasting the \$80.7 billion direct public cost against the \$1 trillion comprehensive social and economic burden.

7.1 Loss Category: Friction (Tax Compliance)

Americans spend 7.1-7.9 billion hours annually complying with the tax code¹⁴⁰.

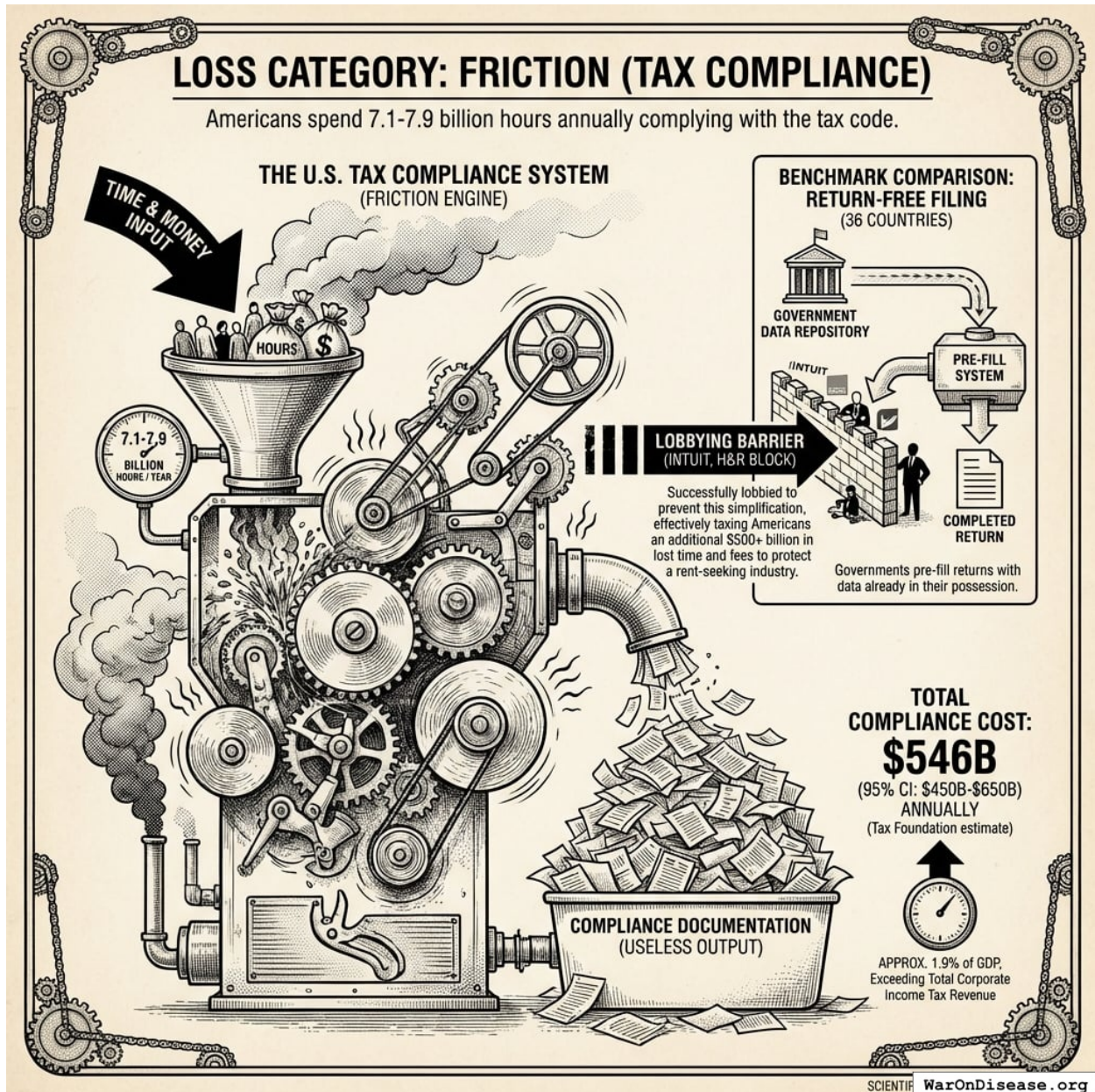


Figure 23: A comparison showing that U.S. tax compliance costs (\$546 billion) exceed total corporate income tax revenue, representing 1.9% of national GDP.

Total compliance cost: \$546B (95% CI: \$450B-\$650B) annually (Tax Foundation estimate)^{16,140}

This is approximately 1.9% of GDP, exceeding total corporate income tax revenue. The labor produces nothing but compliance documentation.

Benchmark comparison: Thirty-six countries use “Return-Free Filing” where governments pre-fill returns with data already in their possession¹⁷⁰. The U.S. tax preparation lobby (Intuit, H&R Block) has successfully lobbied to prevent this simplification¹⁷¹, effectively taxing Americans an additional \$500+ billion in lost time and fees to protect a rent-seeking industry.

7.2 Loss Category: Friction (Housing/Zoning Misallocation)

Local zoning regulations artificially restrict housing supply in high-productivity cities, preventing labor mobility to productive clusters.

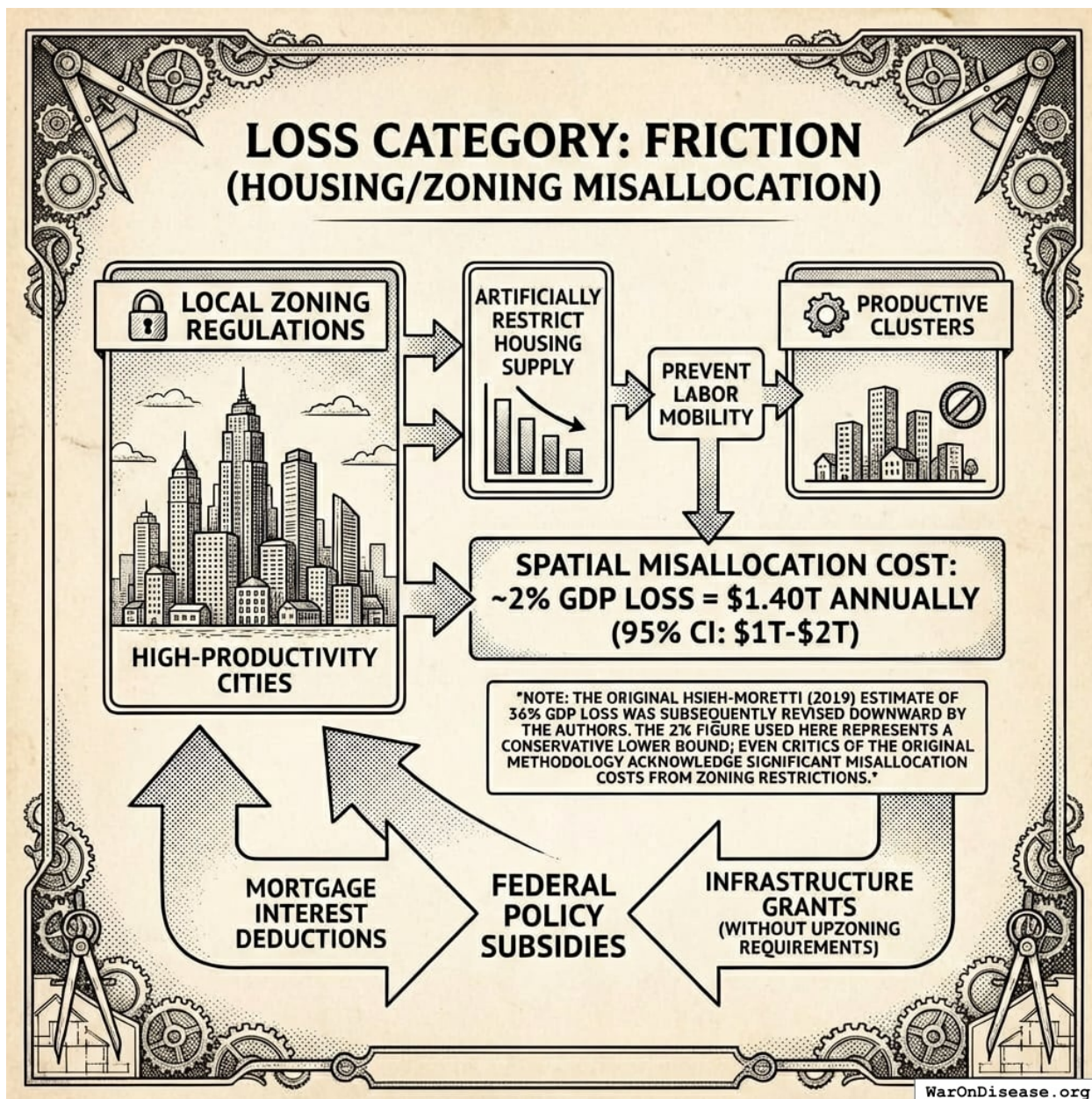


Figure 24: A visualization of the annual GDP loss attributed to housing misallocation, showing a central estimate of \$1.40 trillion within a \$1 trillion to \$2 trillion confidence interval range.

Spatial misallocation cost: Even conservative estimates point to 2% GDP loss = \$1.40T (95% CI: \$1T-\$2T) annually^{172,173}.

Note: The original Hsieh-Moretti (2019) estimate of 36% GDP loss was subsequently revised downward by the authors. The 2% figure used here represents a conservative lower bound; even

critics of the original methodology acknowledge significant misallocation costs from zoning restrictions. Federal policy subsidizes this dysfunction via mortgage interest deductions and infrastructure grants without upzoning requirements.

7.3 Loss Category: Idle (NEPA Permitting Delays)

The National Environmental Policy Act forces infrastructure projects into multi-year review. Average Environmental Impact Statement: 4.5 years.



Figure 25: A visualization illustrating the 4.5-year average NEPA review timeline and the resulting \$100-140 billion in annual economic losses, highlighting the disproportionate impact on clean energy projects.

Delay costs: \$100-140 billion annually in lost returns and capital efficiency¹⁷⁴.

NEPA creates a “Green Paradox”: delaying clean energy projects (transmission, wind, geothermal) more than fossil fuel projects, undermining stated policy goals.

7.4 Loss Category: Parasitic (Jones Act)

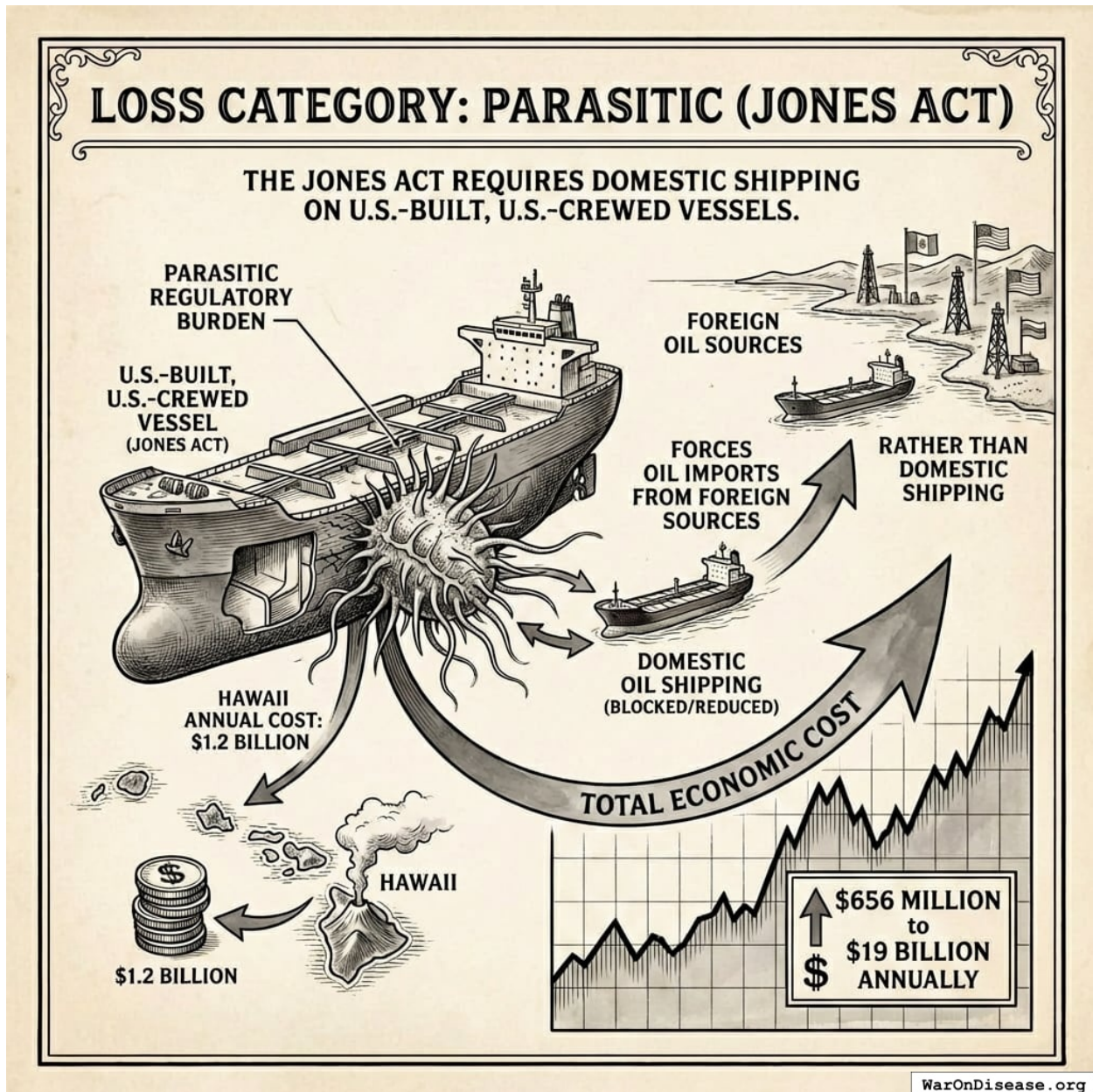


Figure 26: A comparison chart showing the \$1.2 billion annual cost to Hawaii relative to the estimated national economic impact of the Jones Act, which ranges from \$656 million to \$19 billion.

The Jones Act requires domestic shipping on U.S.-built, U.S.-crewed vessels. Results:

- Hawaii annual cost: \$1.2 billion¹⁷⁵

- Forces oil imports from foreign sources rather than domestic shipping
- Total economic cost: \$656 million to \$19 billion annually¹⁷⁶

7.5 Regulatory Subsystem Summary

Loss Category	Low Estimate	Mean	High Estimate
Friction (tax compliance)	\$464B	\$505B	\$546B
Friction (housing)	\$300B	\$560B	\$700B
Idle (NEPA)	\$100B	\$120B	\$140B
Parasitic (Jones Act)	\$1B	\$10B	\$19B
Total	\$600B	\$900B	\$1.2T

8 Subsystem Audit: Subsidies and Transfers

Direct transfers to profitable industries distort market signals and insulate incumbents from innovation pressure.

8.1 Loss Category: Parasitic (Fossil Fuel Subsidies)

Direct annual subsidies to fossil fuel companies: \$10-52 billion¹⁷⁷.

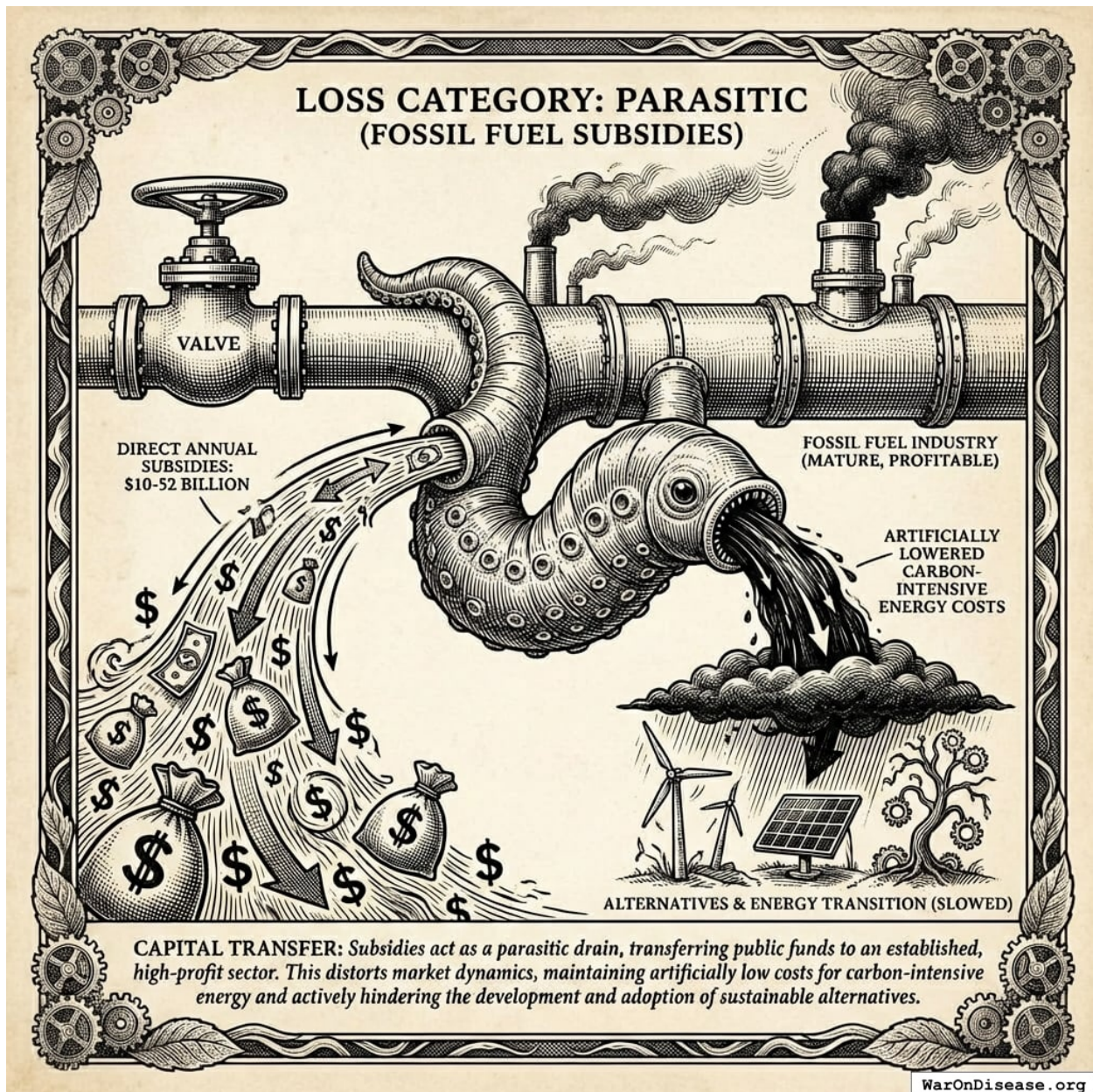


Figure 27: A diagram illustrating the flow of fossil fuel subsidies and their resulting impact on market prices and the pace of the energy transition.

This represents capital transfer to a mature, profitable industry, artificially lowering carbon-intensive energy costs relative to alternatives and slowing energy transition.

8.2 Loss Category: Parasitic (Agricultural Subsidies)

Agricultural subsidies in 2024: \$9.3-30 billion¹⁷⁸. Distribution is regressive. Top 10% of recipients received 65% of payments in 2024¹⁷⁹.

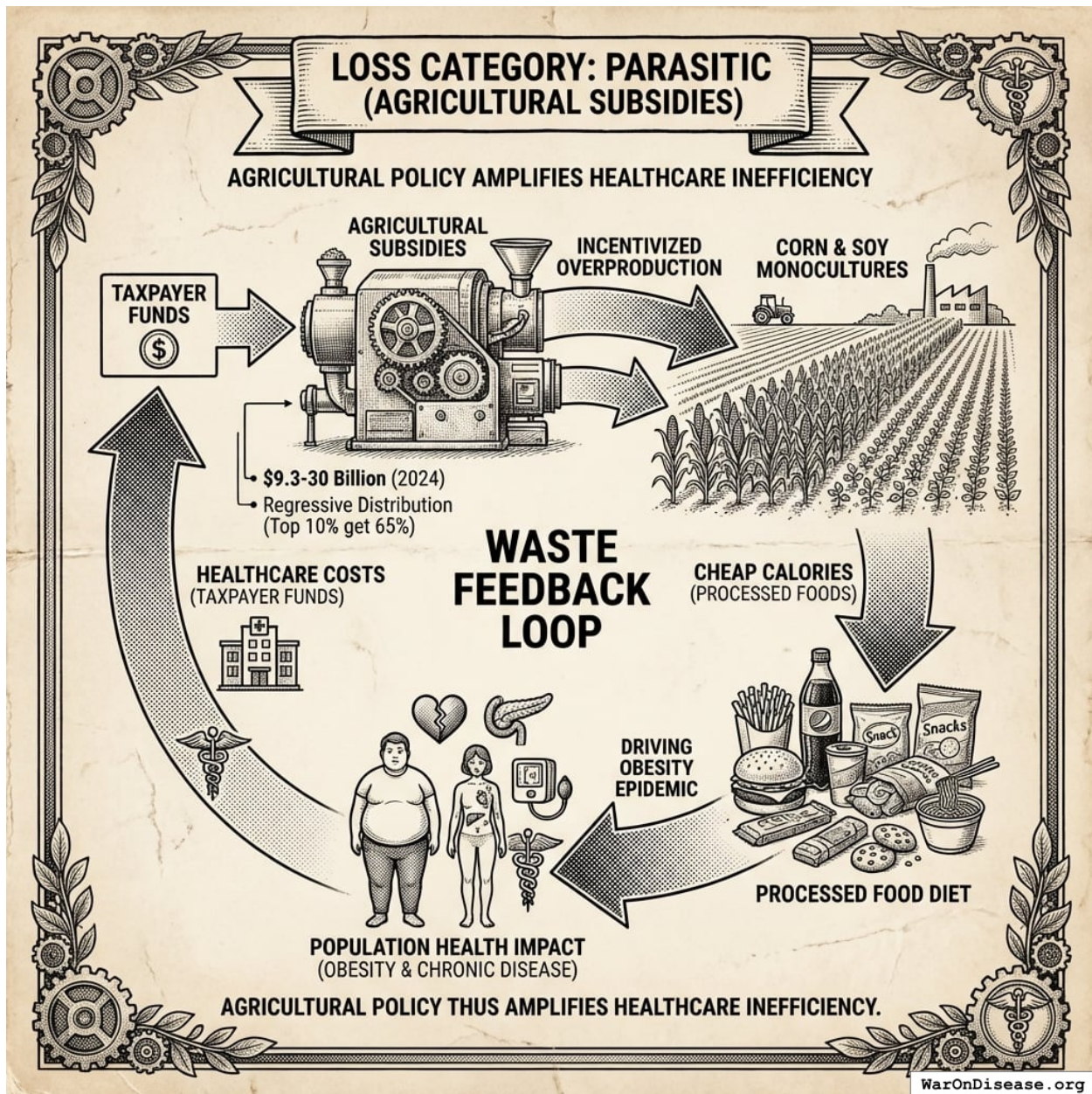


Figure 28: A flowchart depicting the ‘Waste Feedback Loop’ where regressive agricultural subsidies drive the overproduction of corn and soy, leading to a cycle of processed food consumption, obesity, and increased healthcare expenditures.

The subsidy structure incentivizes overproduction of corn and soy, which form the backbone of the processed food diet driving the obesity epidemic. This creates a **Waste Feedback Loop**: taxpayer funds subsidize production of cheap calories that make the population sick, requiring additional taxpayer funds to treat the resulting chronic disease (see Healthcare subsystem). Agricultural policy thus amplifies healthcare inefficiency.

8.3 Loss Category: Negative Work (Tariffs and Corporate Welfare)

Corporate welfare: Cato Institute tallies \$181 billion annually in grants, loans, and credits to specific businesses¹⁸⁰.

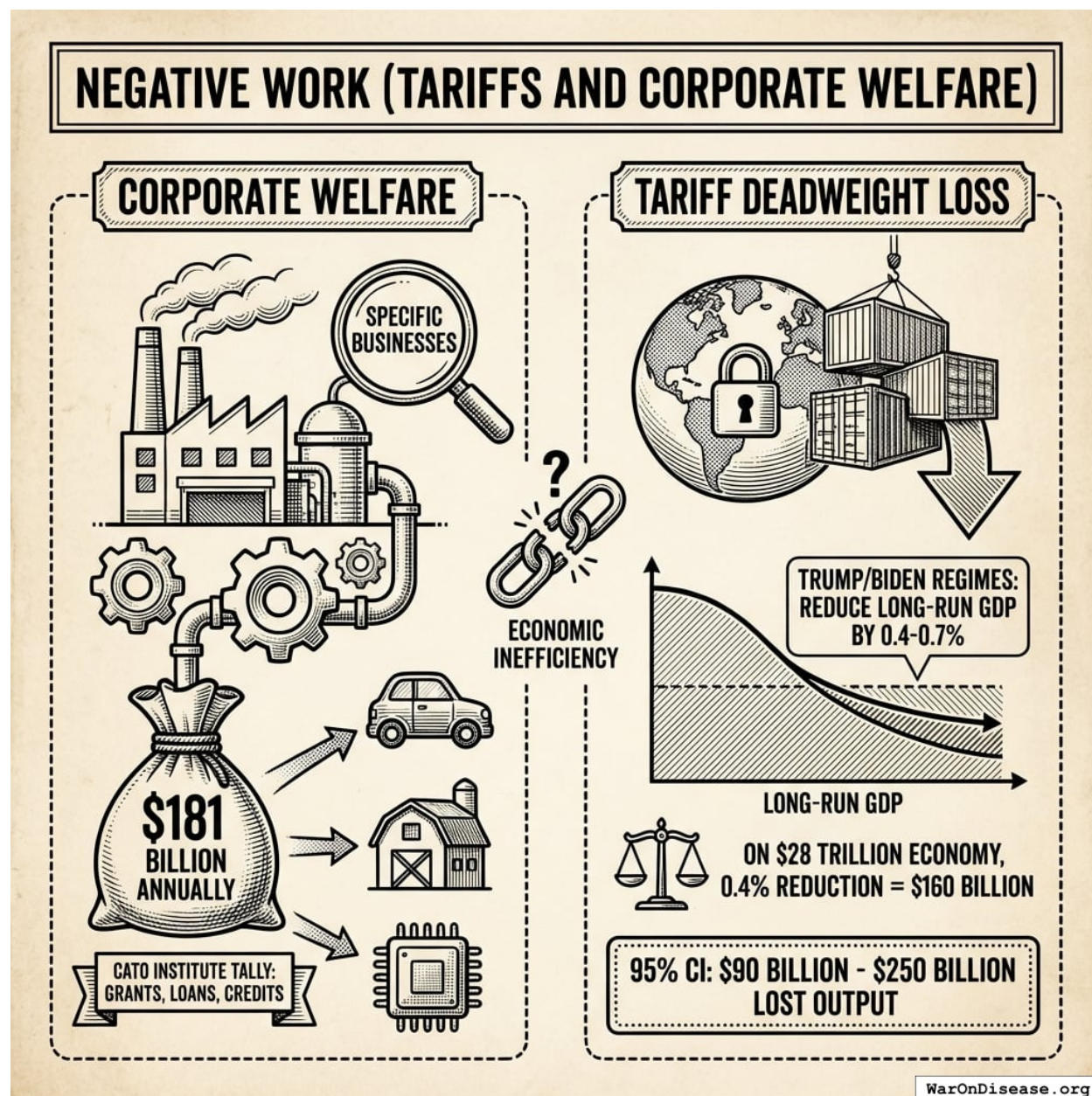


Figure 29: A comparison of the annual economic costs of corporate welfare and tariff deadweight loss, highlighting the similar scale of impact between \$181 billion in subsidies and the \$160 billion loss in output.

Tariff deadweight loss: Trump/Biden tariff regimes estimated to reduce long-run GDP by 0.4-0.7%¹⁸¹. On a \$28 trillion economy, 0.4% reduction = \$160B (95% CI: \$90B-\$250B) in lost output.

8.4 Subsidies Subsystem Summary

Loss Category	Low Estimate	Mean	High Estimate
Parasitic (fossil fuel)	\$10B	\$30B	\$52B
Parasitic (agriculture)	\$9B	\$20B	\$30B
Parasitic (corporate)	\$150B	\$181B	\$200B
Negative work (tariffs)	\$80B	\$112B	\$150B
Total	\$200B	\$280B	\$350B

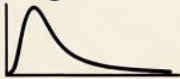
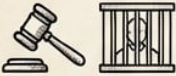
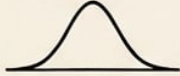
9 Aggregate Efficiency Calculation

9.1 Monte Carlo Simulation Parameters

We model Total Efficiency Gap as the sum of subsystem losses, using Monte Carlo simulation to account for correlated uncertainties.

MONTE CARLO SIMULATION PARAMETERS

We model Total Efficiency Gap as the sum of subsystem losses, using Monte Carlo simulation to account for correlated uncertainties.

Subsystem	Distribution	P10	Mean	P90	Rationale
	Log-Normal 	\$100B	\$210B	\$350B	Audit opacity creates high upper-bound risk
	Normal 	\$300B	\$450B	\$600B	OECD comparisons provide stable variance
	Normal 	\$150B	\$280B	\$400B	Incarceration costs relatively predictable
	Log-Normal 	\$600B	\$900B	\$1.5T	Housing/tax opportunity costs have fat tails
	Normal 	\$200B	\$280B	\$350B	Direct line items easier to bound
	Constant 	\$150B	\$150B	\$150B	Interest on accumulated inefficiency

Note: Debt interest calculated as portion of net interest (\$888B in 2024) attributable to cumulative efficiency gap.

WarOnDisease.org

Figure 30: A range chart comparing the mean annual efficiency gap across six subsystems, with error bars representing the P10 to P90 uncertainty intervals.

Subsystem	Distribution	P10	Mean	P90	Rationale
Defense	Log-Normal	\$100B	\$210B	\$350B	Audit opacity creates high upper-bound risk
Healthcare	Normal	\$300B	\$450B	\$600B	OECD comparisons provide stable variance

Subsystem	Distribution	P10	Mean	P90	Rationale
Justice/Prohibition	Normal	\$150B	\$280B	\$400B	Incarceration costs relatively predictable
Regulatory/Tax	Log-Normal	\$600B	\$900B	\$1.5T	Housing/tax opportunity costs have fat tails
Subsidies	Normal	\$200B	\$280B	\$350B	Direct line items easier to bound
Debt Interest	Constant	\$150B	\$150B	\$150B	Interest on accumulated inefficiency

Note: Debt interest calculated as portion of net interest (\$888B in 2024)¹⁸² attributable to cumulative efficiency gap.

9.2 Simulation Results

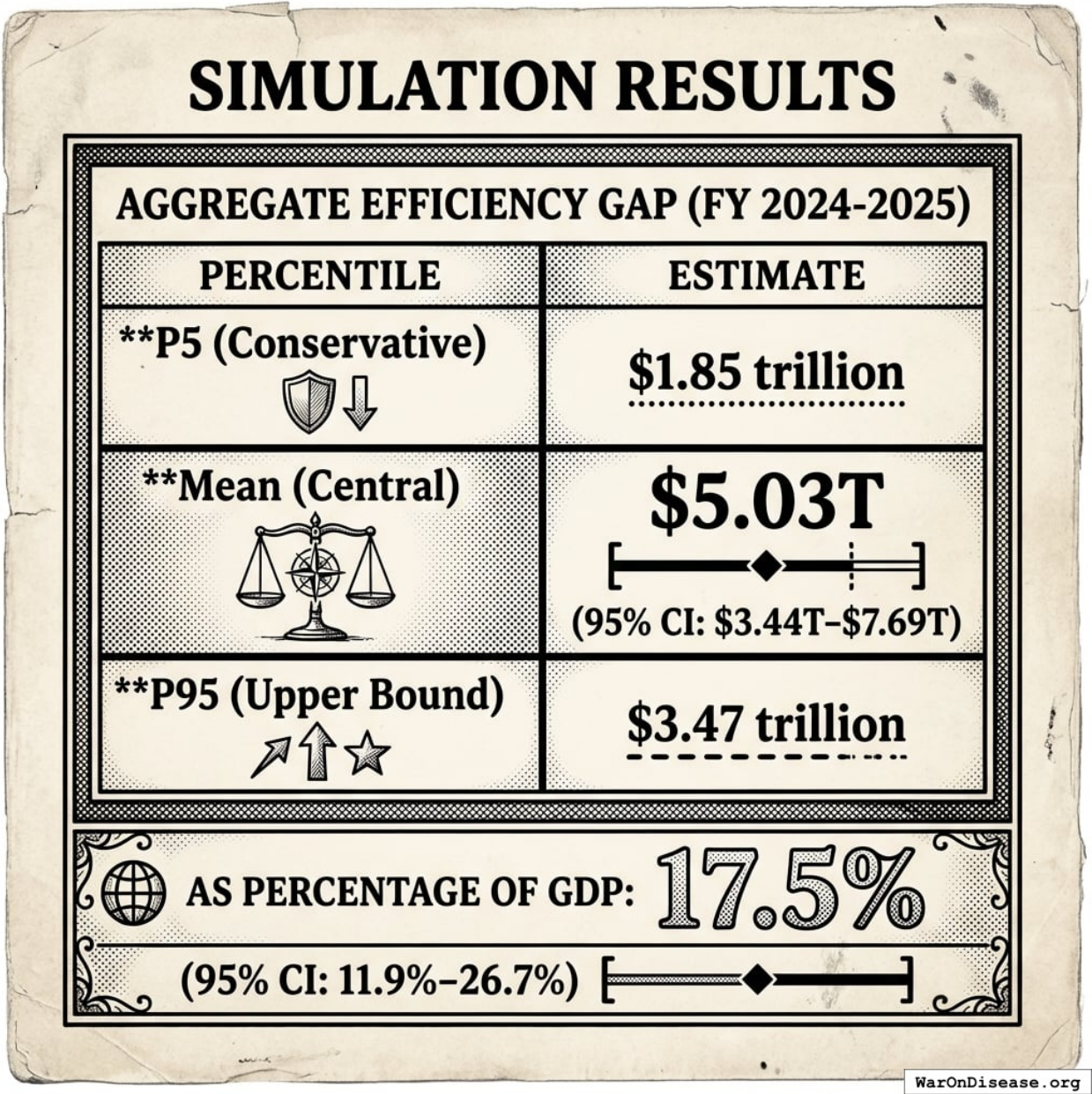


Figure 31: A comparison chart of the FY 2024-2025 Aggregate Efficiency Gap estimates, illustrating the mean value of \$5.03 trillion alongside its 95% confidence interval and percentile distributions.

Aggregate Efficiency Gap (FY 2024-2025):

Percentile	Estimate
P5 (Conservative)	\$1.85 trillion
Mean (Central)	\$5.03T (95% CI: \$3.44T-\$7.69T)
P95 (Upper Bound)	\$3.47 trillion

As percentage of GDP: 17.5% (95% CI: 11.9%-26.7%)

9.2.1 Subsystem Uncertainty Distributions

The following figures show Monte Carlo distributions for key subsystem loss estimates:

10 Tax Compliance

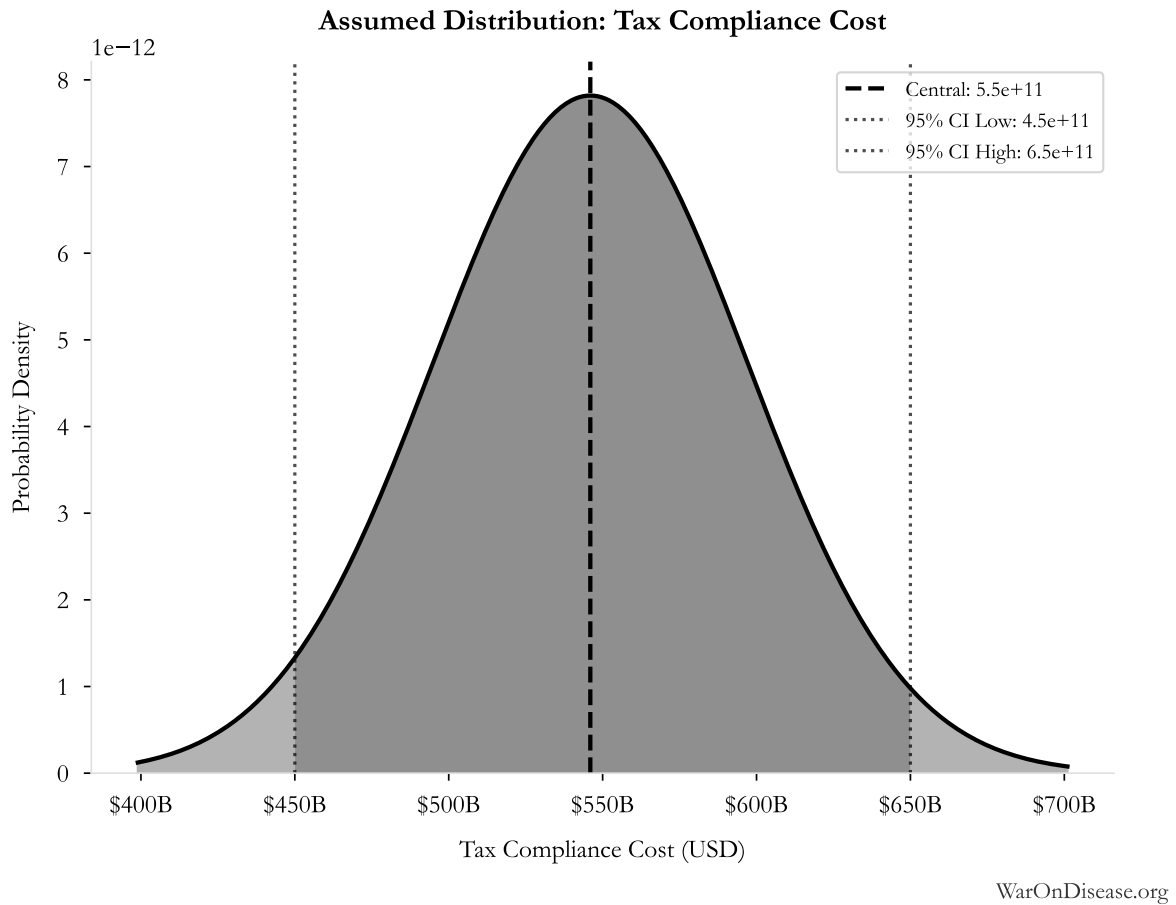


Figure 32: Probability Distribution: Tax Compliance Cost

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.

11 Housing/Zoning

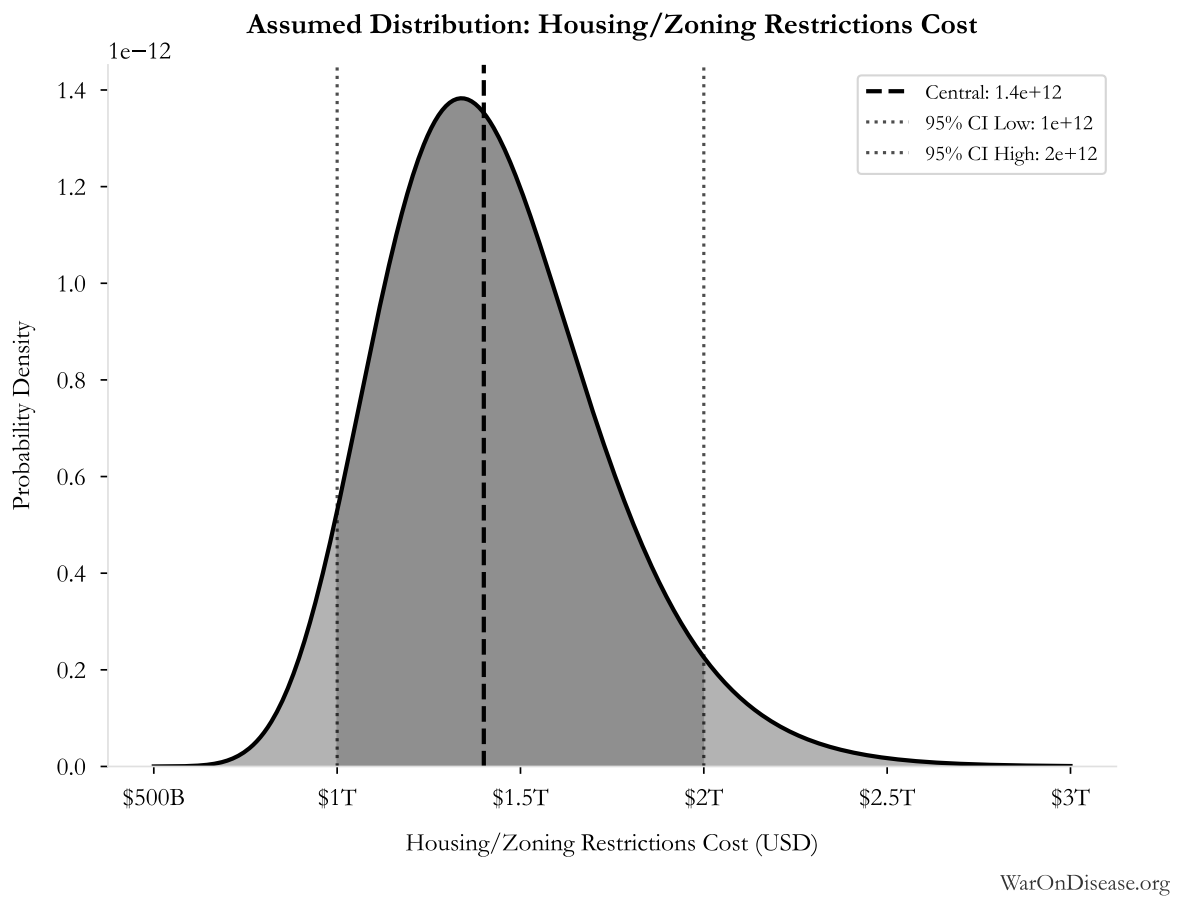


Figure 33: Probability Distribution: Housing/Zoning Restrictions Cost

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.

12 Healthcare

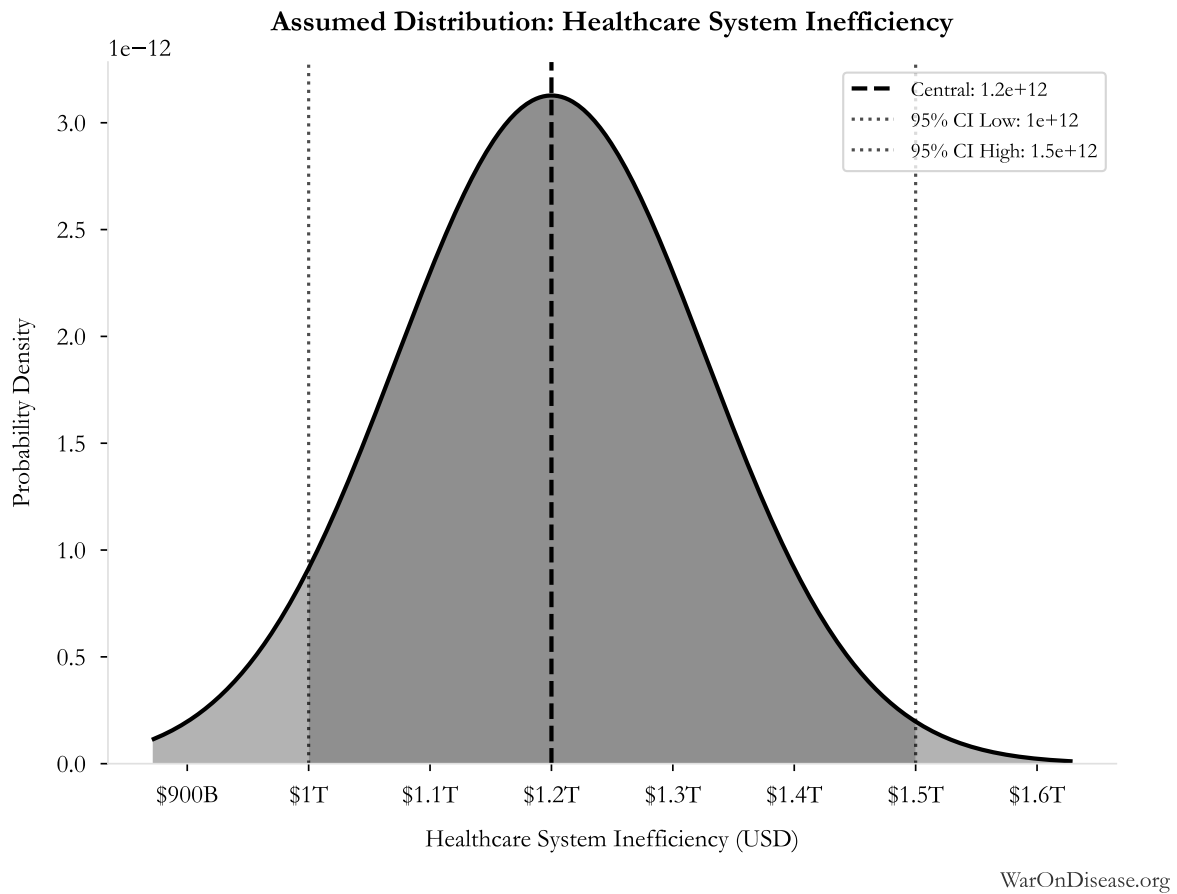


Figure 34: Probability Distribution: Healthcare System Inefficiency

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.

13 Drug War

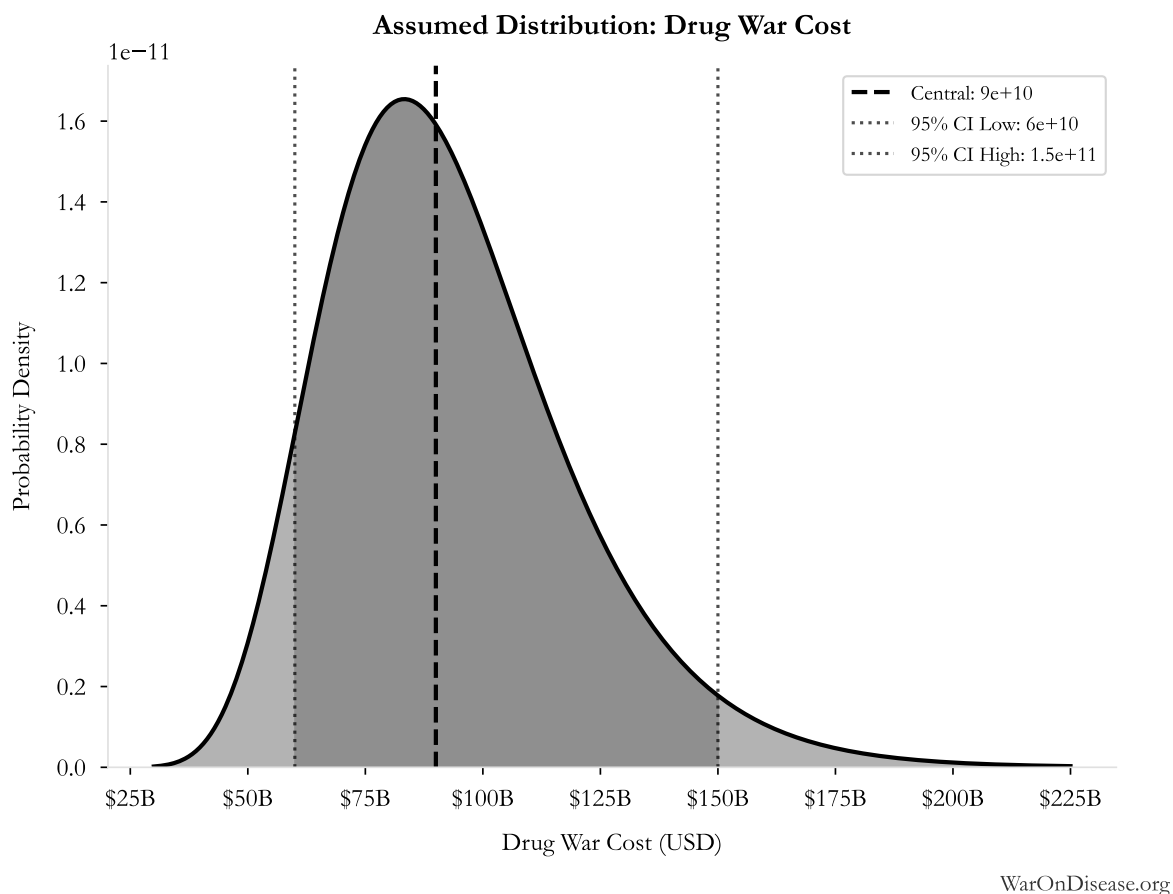


Figure 35: Probability Distribution: Drug War Cost

This chart shows the assumed probability distribution for this parameter. The shaded region represents the 95% confidence interval where we expect the true value to fall.

13.1 Efficiency Rating Calculation

With system input of ~\$6.75 trillion and efficiency gap of \$2.27-3.47 trillion:

$$\text{Efficiency} = \frac{\text{Input} - \text{Gap}}{\text{Input}} = \frac{6.75 - 2.27}{6.75} = 66\% \text{ (mean)}$$

Range: 52-67% depending on gap estimate used.

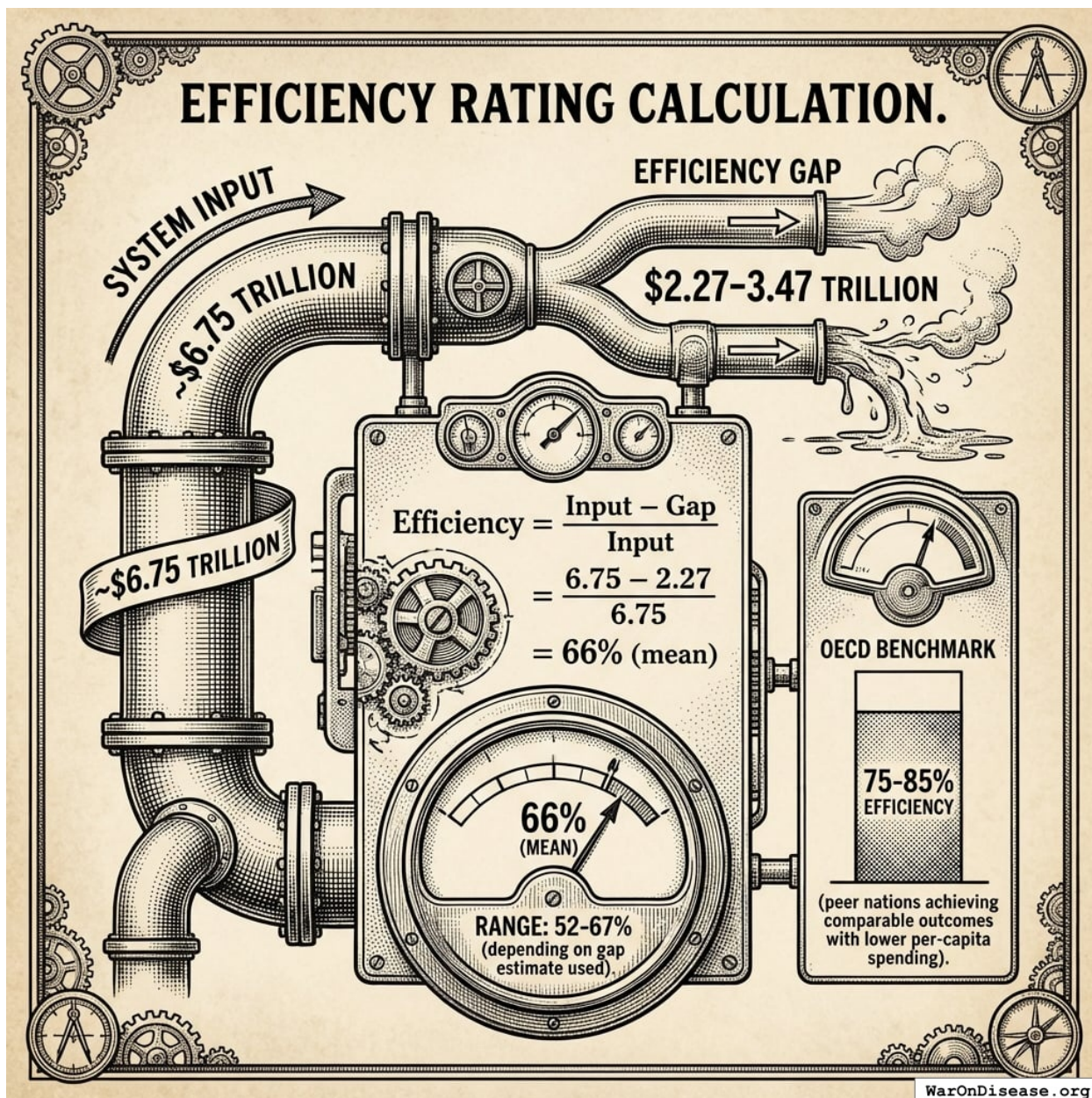


Figure 36: A comparison chart showing the current efficiency range and mean value relative to the significantly higher OECD benchmark range.

OECD benchmark: 75-85% efficiency (peer nations achieving comparable outcomes with lower per-capita spending).

13.2 Human Cost Quantification (Economic Equivalents)

To contextualize the efficiency gap in human terms, we apply standard valuation thresholds. **These are economic equivalents, not epidemiological mortality counts.**

**Using VSL (13.7million) : **VSL-Equivalents = $\frac{\$2.27 \text{ trillion}}{\$13.7 \text{ million}} \approx 165,700\$$

**Using QALY threshold (100,000) : **QALY-Equivalents = $\frac{\$2.27 \text{ trillion}}{\$100,000} = 22.7 \text{ million}$

Interpretation: The efficiency gap represents foregone welfare equivalent to 22.7 million quality-adjusted life years annually. This does not mean 165,700 people die from inefficiency. Rather, the misallocated resources *could have* purchased health improvements of that magnitude if deployed at cost-effectiveness thresholds used in medical decision-making.

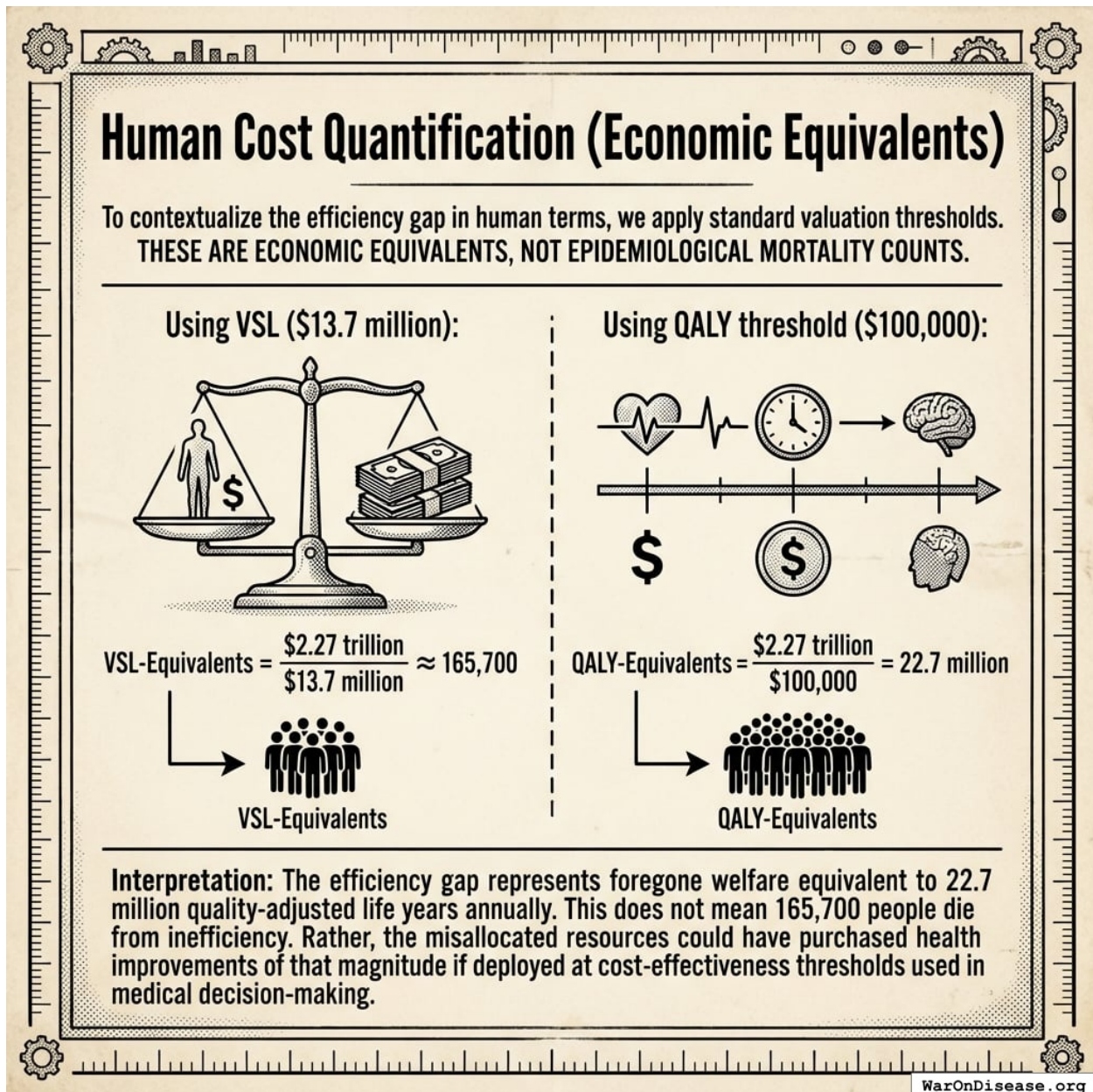


Figure 37: A comparison showing the \$2.27 trillion efficiency gap converted into human cost equivalents: 165,700 value-of-statistical-life (VSL) units and 22.7 million quality-adjusted life years (QALYs).

14 Reallocation Potential

If U.S. efficiency improved to OECD median (80%), approximately **\$1.5 trillion annually** becomes available for reallocation.

14.1 Context Comparisons

Initiative	Annual Cost	Efficiency Gap Coverage
1% Treaty funding	\$27.2B	55x covered
Global disease R&D (current)	\$150B ¹⁸³	Would 10x
U.S. infrastructure backlog	\$2.6T total ¹⁸⁴	Covered in <2 years
Global poverty elimination	~\$175B ¹⁸⁵	8x covered
Complete grid decarbonization	\$100B/year ¹⁸⁶	15x covered

The efficiency gap is not abstract accounting. It represents real capacity currently unavailable for health, infrastructure, and security improvements.

15 Structural Factors

Why do these losses persist despite apparent obviousness? Several structural factors explain system inertia:

15.1 Severed Feedback Loops

Government programs lack market feedback mechanisms. A private firm losing \$210 billion annually on inefficient logistics would face bankruptcy. Federal agencies face no equivalent selection pressure.

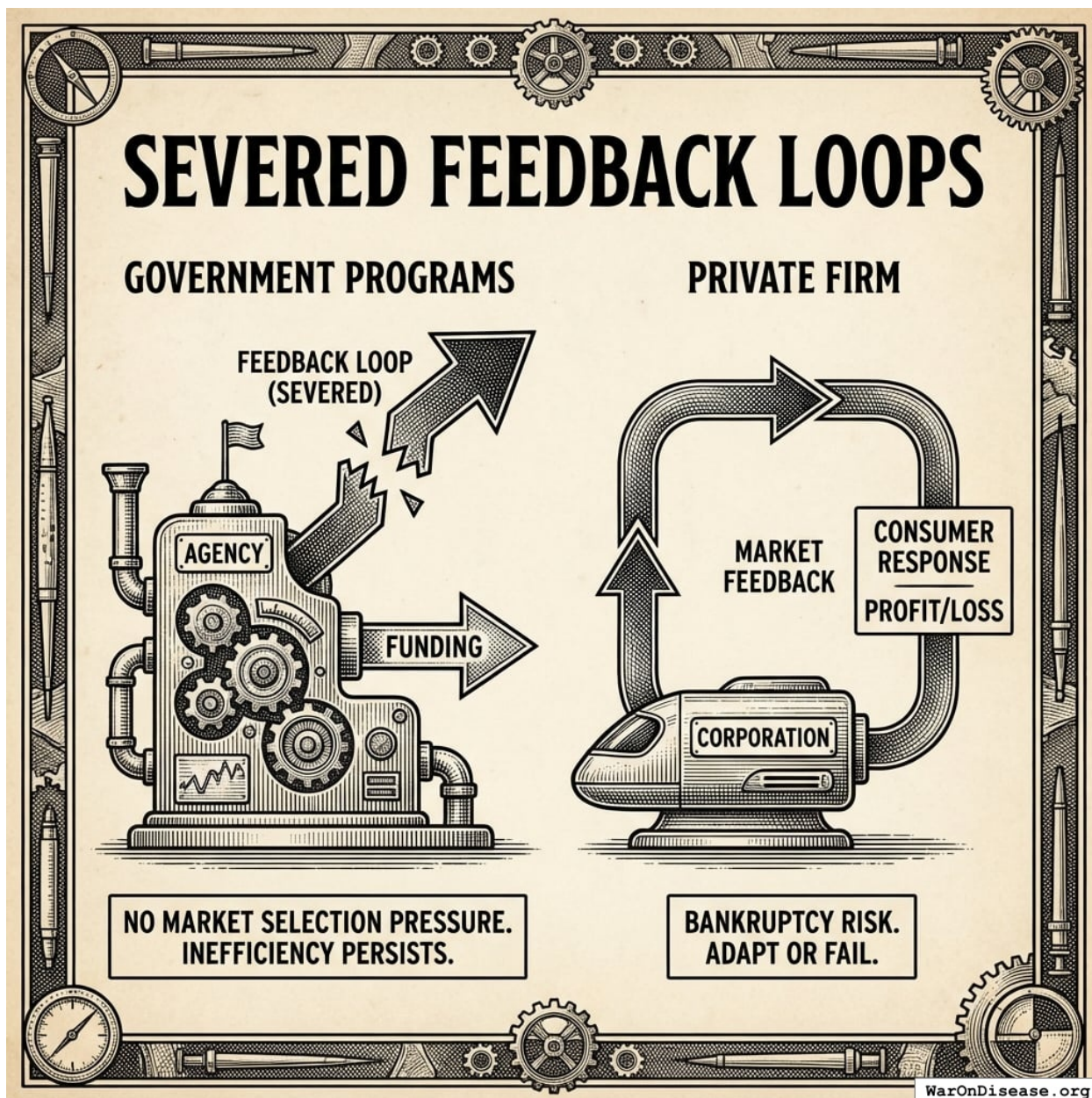


Figure 38: A comparison showing the closed feedback loop of a private firm where losses lead to bankruptcy versus the severed feedback loop of a federal agency lacking market selection pressure.

15.2 Principal-Agent Misalignment

Those administering programs (bureaucrats, contractors) have incentives misaligned with program objectives. Contractors profit from complexity; administrators expand headcount regardless of output.

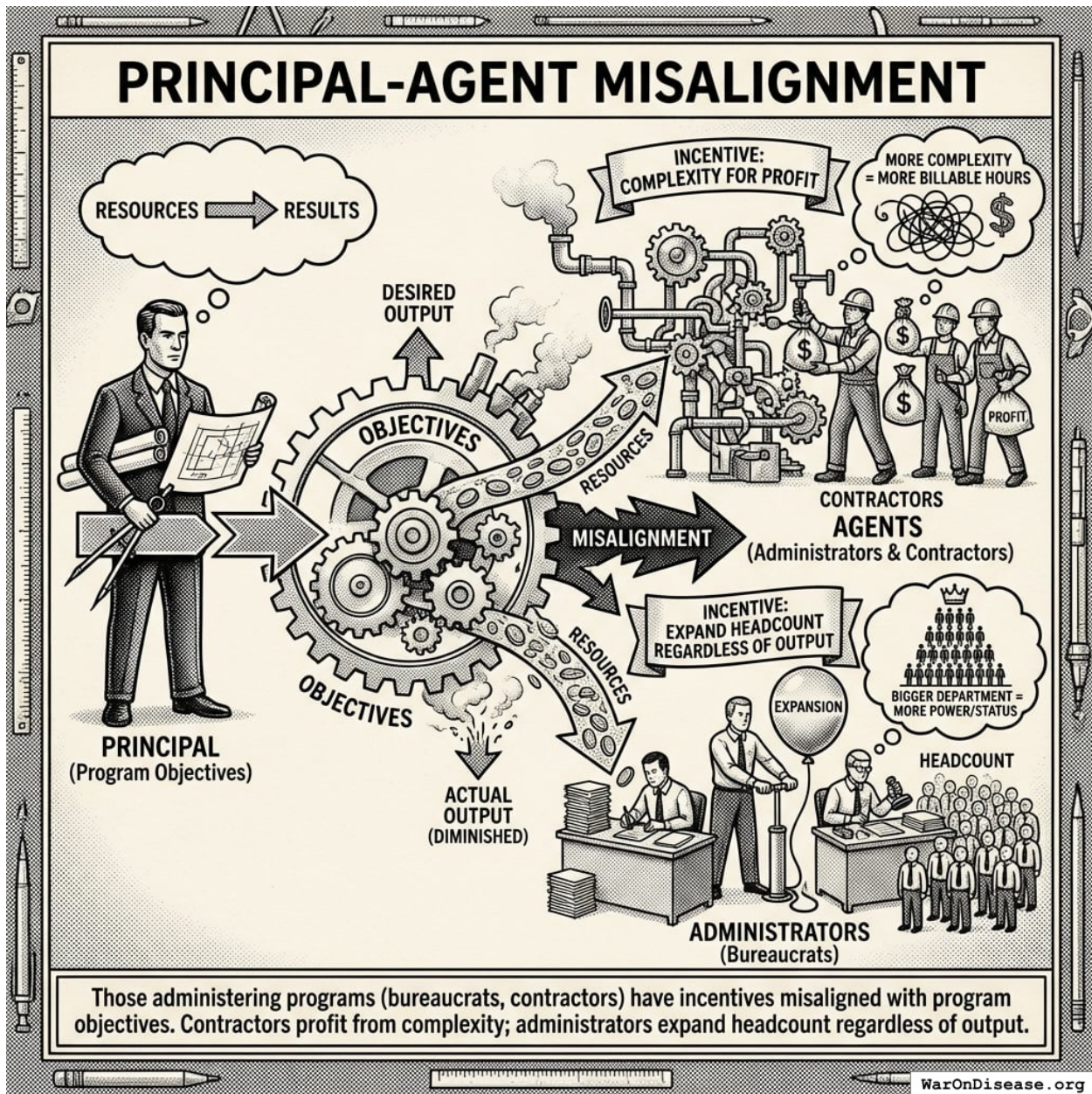


Figure 39: A conceptual diagram illustrating the divergence between program objectives and the individual incentives of bureaucrats and contractors.

15.3 Measurement Failure

Current accounting measures expenditure, not utility. A dollar spent equals a dollar of “activity” regardless of outcome. Without output measurement, optimization is impossible.

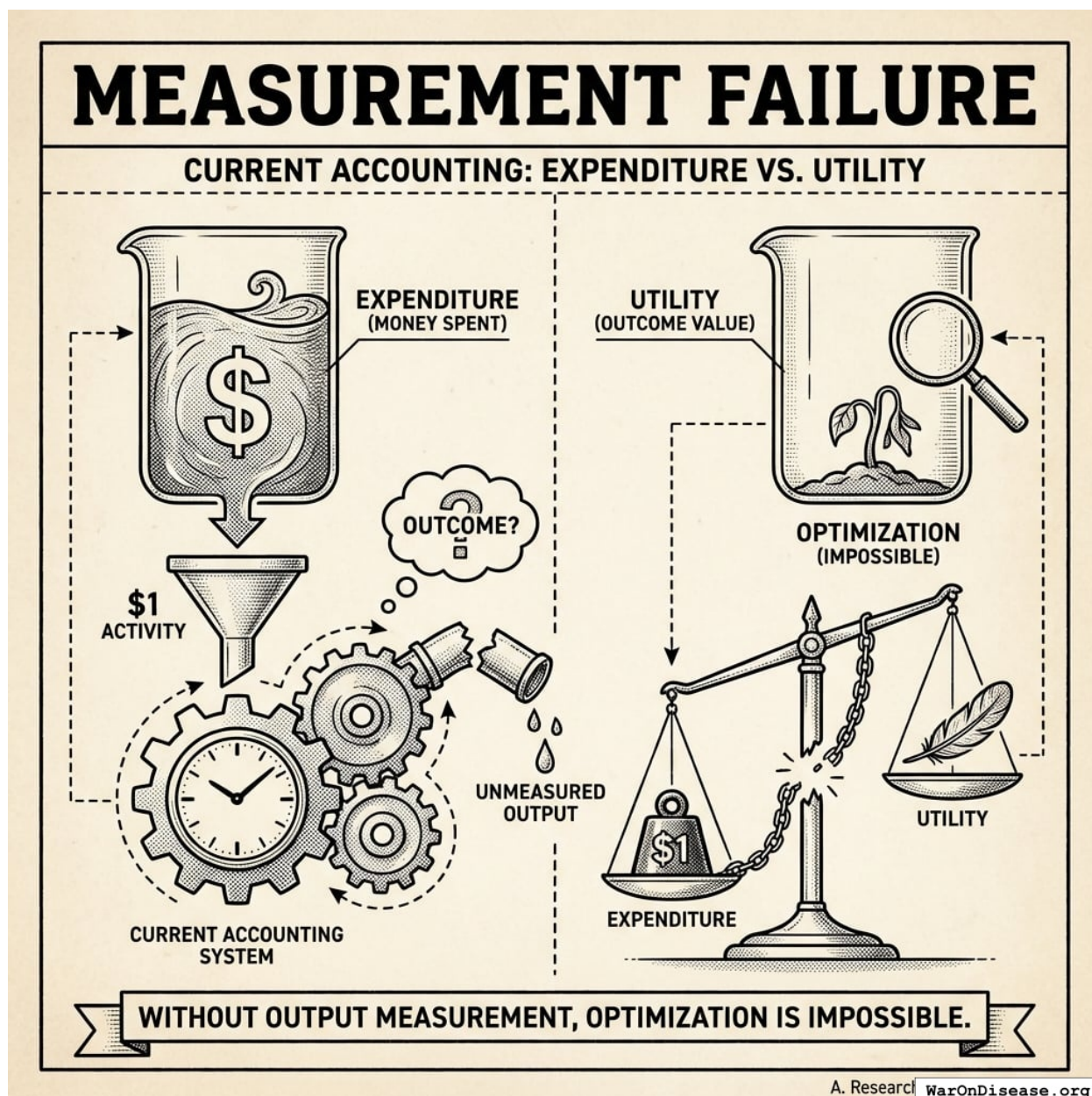


Figure 40: A comparison between the current accounting model, which equates expenditure directly with activity, and an optimized model that focuses on measuring utility and outcomes.

15.4 Monopoly Dynamics

Government services typically face no competition. Without competitive pressure, innovation lags and costs inflate. This is the standard monopoly outcome.

15.5 Time Horizon Mismatch

Political cycles (2-4 years) misalign with infrastructure and policy cycles (10-30 years). Long-term efficiency investments lose to short-term visible spending.

16 Confidence Intervals and Limitations

16.1 Estimate Confidence by Subsystem

Subsystem	Data Quality	Confidence
Healthcare Admin	High (OECD comparisons)	High
Tax Compliance	High (IRS data)	High
Defense Audit	Low (61% unaccounted)	Medium
Incarceration	Medium (direct costs clear, indirect estimated)	Medium
Housing Misallocation	Medium (model-dependent)	Medium
Drug War Opportunity Cost	Low (counterfactual)	Low

16.2 What This Analysis Excludes

- State/local inefficiency beyond federal mandates
- Implicit subsidies (unpriced externalities)
- Intergenerational costs (debt burden on future)
- Second-order behavioral effects
- International competitiveness losses

Including these factors would increase the efficiency gap estimate substantially.

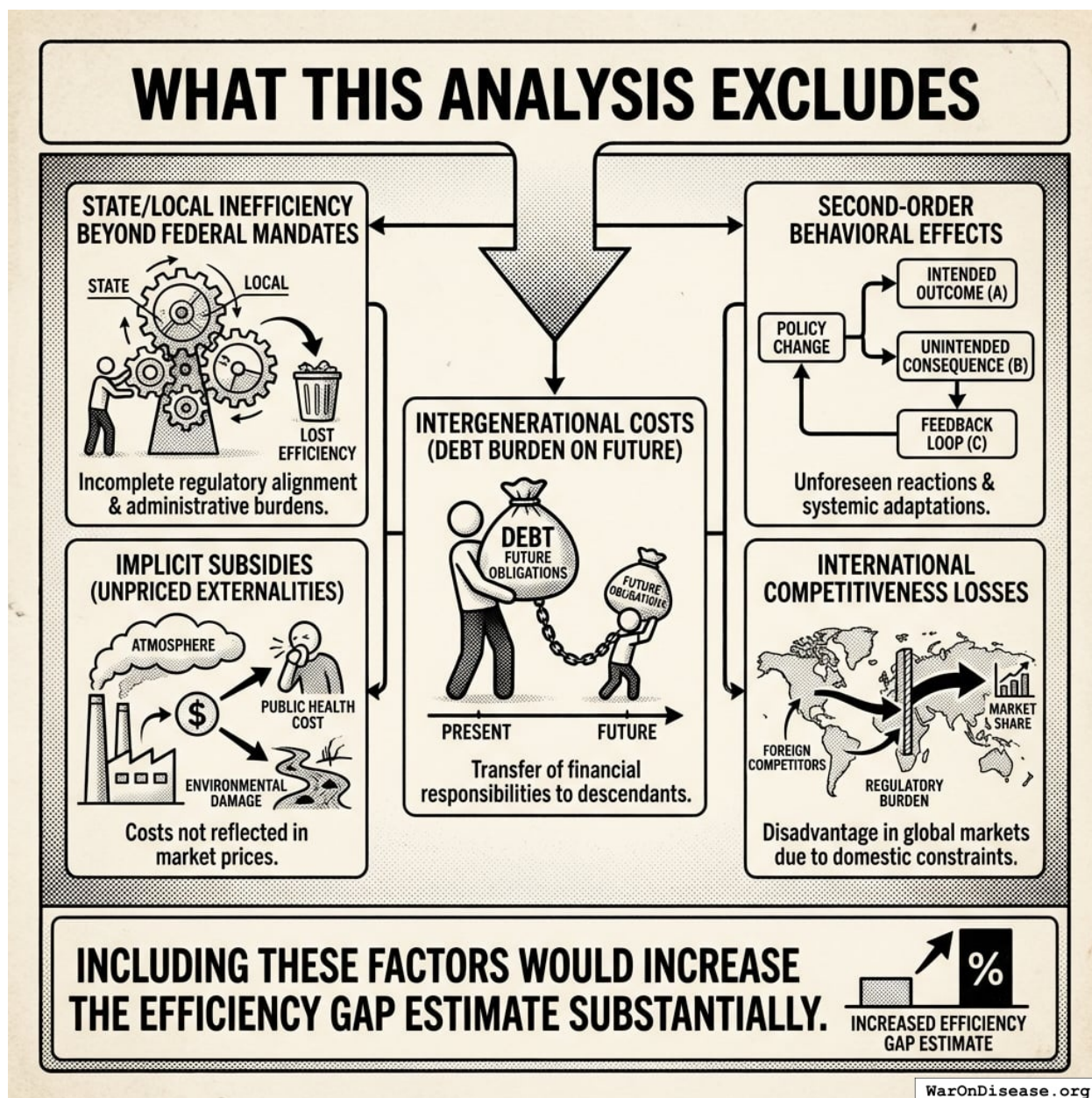


Figure 41: A visualization showing the current efficiency gap estimate compared to the significantly larger gap that would exist if excluded factors like implicit subsidies and intergenerational costs were included.

16.3 Methodological Limitations

1. **Counterfactual uncertainty:** Some estimates require modeling what “would have happened” under alternative policies
2. **Attribution challenges:** Separating federal from state/local effects
3. **Valuation debates:** VSL and QALY thresholds vary by methodology
4. **Data opacity:** DoD audit failures mean some estimates are necessarily imprecise

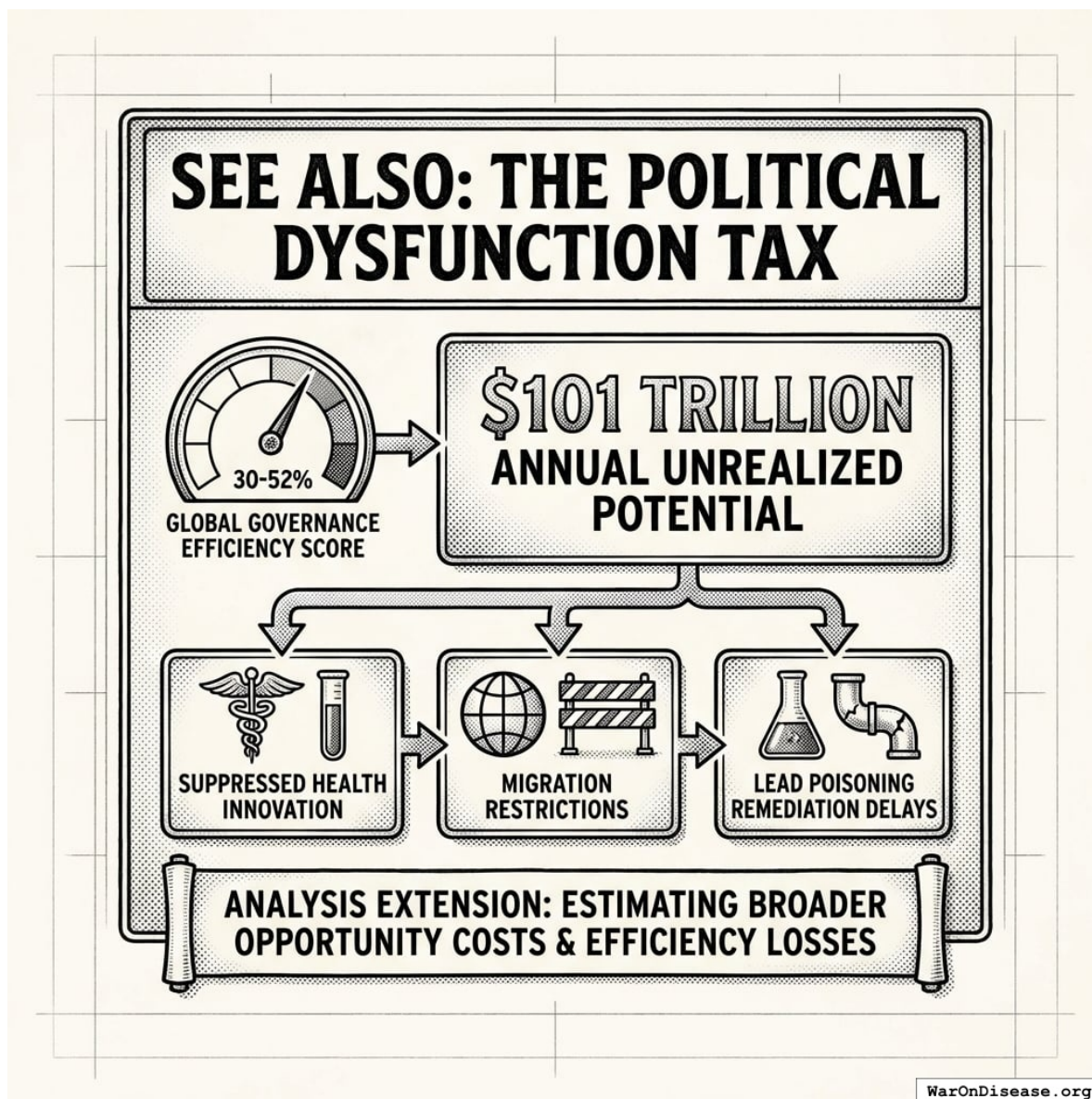


Figure 42: A visualization showing the 30-52% Global Governance Efficiency range and the \$101 trillion in annual unrealized potential across sectors like health, migration, and environmental remediation.

For global perspective on governance efficiency and broader opportunity costs of political dysfunction, see [The Political Dysfunction Tax](#), which extends this analysis to estimate a Global Governance Efficiency Score of 30-52% and identifies \$101 trillion in annual unrealized potential from suppressed health innovation, migration restrictions, and lead poisoning remediation delays.

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Berkshire’s compounded annual return from 1965 through 2024 was 19.9%, nearly double the 10.4% recorded by the S&P 500. Berkshire shares skyrocketed 5,502,284% compared to the S&P 500’s 39,054% rise during that period. Additional sources: <https://www.cnbc.com/2025/05/05/warren-buffetts-return-tally-after-60-years-5502284percent.html> | <https://www.slickcharts.com/berkshire-hathaway/returns>
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